



Advances and Challenges in Constructing Bone Organoids Using Cells Derived from Human Pluripotent Stem Cells: A Review

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Received: 14 August 2025 / Accepted: 28 October 2025

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Abstract

Bone organoids mimic the structure and function of actual bone tissue and serve as novel tools for disease modeling, drug testing, and bone repair. However, their development is severely impeded by the limited availability of cell sources. Fortunately, human pluripotent stem cells (hPSCs) can differentiate into organoid constituent cells, including osteoblasts, osteoclasts, and endothelial cells. However, their differentiation efficiencies are relatively low and do not meet the requirements of clinical applications because of the use of undefined culture components such as fetal bovine serum. More importantly, nearly all the existing scaffolds cannot support the culture of hPSCs. Thus, much effort should be made to construct bone organoids using cells induced from hPSCs. This review starts with the *in vivo* development of bone tissue. We summarize the mechanisms, methods, and purification processes for differentiating hPSCs into the above cell types in bone organoids. On this basis, we described strategies related to hPSC-based bone organoids and the growth factors and bioactive materials needed to accelerate this process. Finally, we extensively discuss the existing challenges and prospects. This review is valuable for the future development and clinical application of hPSC-derived bone organoids.

Keywords Human pluripotent stem cells · Bone organoids · Cell differentiation · Bone tissue engineering

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Background

Natural bone tissue is composed of mineralized bone matrix and vascularized bone marrow chambers while maintaining dynamic equilibrium among osteogenesis (osteoblasts), bone resorption (osteoclasts), and vascularization processes [1, 2]. This complex process poses significant challenges for repairing critical-sized bone defects caused by trauma, tumors, or congenital disorders [3]. While conventional bone tissue engineering strategies—employing seed cells, scaffolds, and bioactive factors—have shown promise in bridging minor defects, they typically rely on single-cell-type approaches and thus fail to replicate the essential multicellular crosstalk and spatial organization of native bone. Consequently, they cannot reproduce the multilayered structure and homeostasis mechanisms of natural bone [4, 5].

Recent advances in bone organoid technology offer a promising strategy to address these limitations. Bone organoids are defined as three-dimensional (3D) *in vitro* micro-tissues that form through the self-organization of multiple cell types or stem cells within a tailored extracellular micro-environment. A key methodological innovation lies in the

co-culture of osteogenic, osteoclastic, and vascular lineages within 3D biomechanically tuned scaffolds, thereby mimicking natural bone tissue and interactions in multicellular crosstalk [6–8]. This bioinspired approach enables the dynamic modeling of complex physiological and pathological processes—such as bone development, homeostasis, and disease progression—under controlled *in vitro* conditions. Therefore, bone organoids have emerged as powerful tools that extend far beyond the capabilities of static biomaterial constructs and enable applications in drug testing, disease modeling, gene manipulation, and regenerative repair.

For many early organoid models—including those for intestine, retina, and brain—human mesenchymal stem cells (hMSCs) and adult stem cells (ASCs) were among the first cell types used [9, 10]. They served as a foundational system in early studies. These hMSC/ASC-based systems have been invaluable for proof-of-concept studies and understanding basic osteoblast behavior within a 3D matrix. However, the translational potential of these models is inherently constrained by the limitations of hMSCs themselves, including donor-dependent heterogeneity, limited expansion capacity, and eventual senescence during *in vitro* passaging [11, 12]. Consequently, achieving large-scale, reproducible bone regeneration with hMSC-derived organoids remains a significant challenge. Furthermore, while these models advanced the field, they often fall short of replicating the complex multicellular crosstalk between osteogenic, osteoclastic, and vascular components, with few studies successfully integrating hard and soft tissues such as mineralized bone and functional bone marrow niches [13, 14]. Fortunately, human pluripotent stem cells (hPSCs), including human embryonic stem cells (hESCs) and human induced pluripotent stem cells (hiPSCs), have the potential for long-term self-renewal and multiple differentiation capacities [15]. All the cell types needed for bone organoids can be induced from hPSCs without immune issues, which also benefits subsequent cell integration for clinical applications [14]. Fig. 1 illustrates the construction of bone organoids using cells derived from human pluripotent stem cells and their applications. It is essential to establish *in vitro* induction systems for the osteogenic, osteoblastic, or vascularized differentiation of hPSCs by adding various inducing factors [16–18]. Unfortunately, Matrigel coatings and animal serum-containing induction media are used for the culture of hPSCs [19]. Establishing an *in vitro* stepwise induction system to obtain a sufficient number of each kind of seed cell under defined and xeno-free conditions remains a considerable challenge [20, 21]. Moreover, problems also exist in the use of cell purification to avoid the formation of teratomas.

The application of various materials, cells, and techniques for creating bone organoids has been the subject of

several reviews in recent years [2, 6, 22, 23]. However, few articles have addressed the intriguing role that hPSCs play in this context or the role that studies of naturally occurring osteogenic developmental processes play in inducing stem cell differentiation *in vitro*. In this review, we first introduce the *in vivo* bone development process and the pathways by which hESCs differentiate into bone tissue constituent cells such as osteoblasts, osteoclasts, and endothelial cells. We then explored the gradual and xeno-free *in vitro* differentiation of hPSCs into the above cells, including the cytokines and signaling pathways involved. We also summarized the reported purification methods used to obtain these differentiated cells. On the basis of these findings, we describe the scaffold materials and construction strategies for bone organoids. Finally, we discuss the existing challenges and prospects, hoping to accelerate the development and application of hPSC-derived bone organoids.

***In Vivo* Development of Bone Tissue and Cellular Components**

To engineer biomimetic bone organoids, it is essential to mirror the cellular composition and spatial organization of natural bone tissue. This necessitates a comprehensive understanding of three fundamental biological processes: (1) the hierarchical organization of the mineralized bone matrix, (2) the processes of natural bone formation, and (3) the ontogeny of functional cell lineages from hESCs, including osteoblasts, osteoclasts, and endothelial cells. Accordingly, in this section, we delineate each step of hESC-to-bone tissue cell differentiation (Fig. 2). This foundational knowledge provides critical references for designing *in vitro* hPSC differentiation strategies that mimic native developmental trajectories, thereby bridging the gap between *in vivo* biology and *in vitro* organoid engineering.

Hierarchical Structures of Bone Tissue

Bone tissue is a complex system with hierarchical structures, integrating the dense outer layers of cortical bone with the porous interior of trabecular bone. These macroscopic structures are further organized into functional subunits: Haversian systems (osteons) form cylindrical conduits for neurovascular bundles, whereas lacunocanalicular networks enable osteocyte communication across mineralized matrices [2]. This hierarchical architecture emerges through the coordinated interplay of three functional units: (1) the hematopoietic unit sustaining marrow microenvironments, (2) the resorption unit remodeling bone through osteoclast activity, and (3) the osteogenic unit depositing a new matrix via osteoblast clusters [24]. This hematopoietic

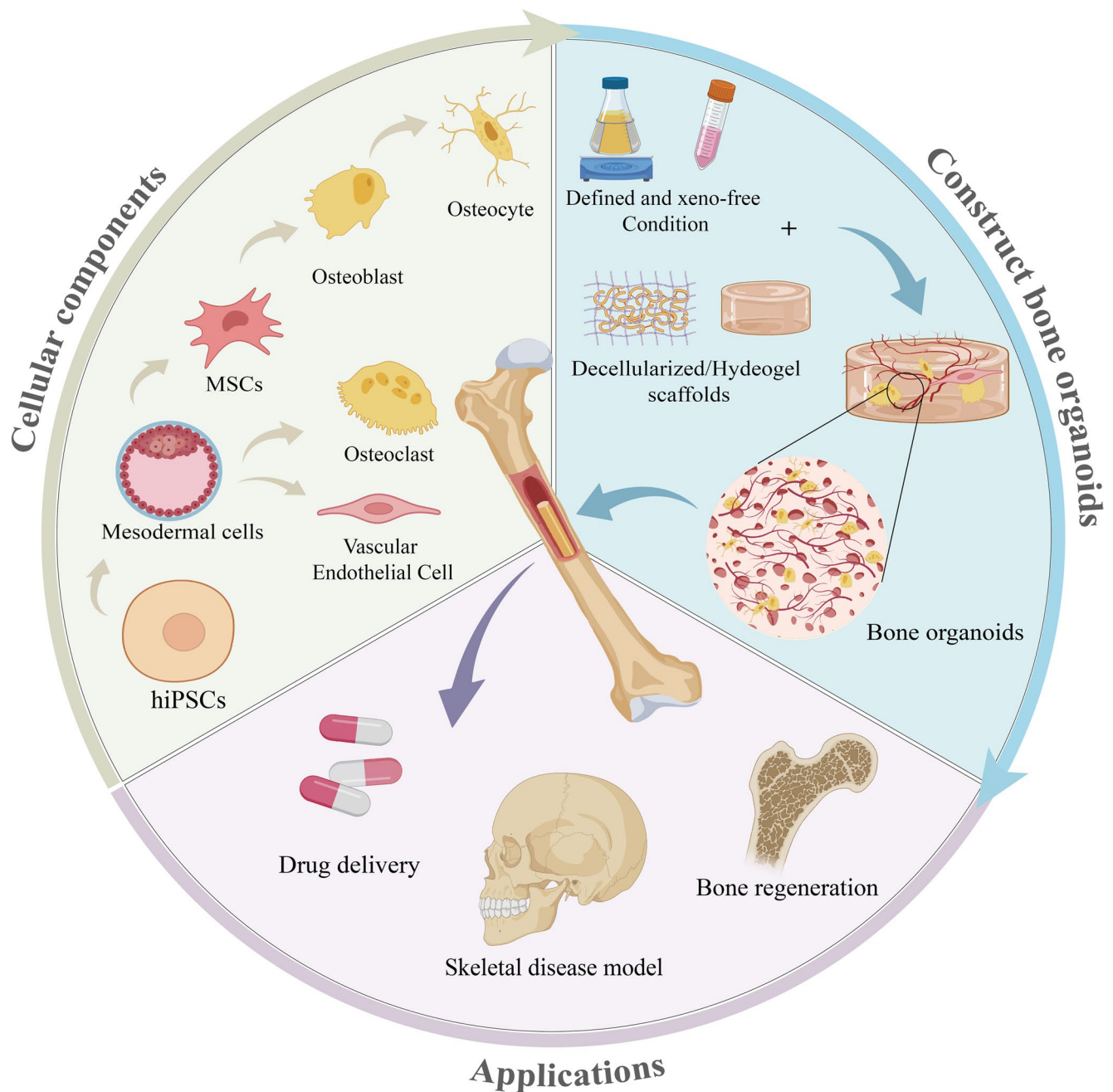


Fig. 1 The construction of bone organoids using cells derived from human pluripotent stem cells and their applications

unit is housed within the bone marrow microenvironment (BMM), a complex and dynamic niche that is critical for maintaining systemic homeostasis. The BMM is not merely a scaffold for blood cell production; it is an integrated system where skeletal and hematopoietic lineages are functionally intertwined. Its complexity arises from a heterogeneous collection of cellular and molecular components. The BMM consists of a diverse array of cells, including hematopoietic stem cells (HSCs), osteoblasts, osteoclasts, adipocytes, endothelial cells, mesenchymal stromal cells, macrophages,

and megakaryocytes. These cells interact through direct contact, secreted factors (e.g., CXCL12, SCF), and extracellular matrix components to regulate HSC maintenance, self-renewal, differentiation, and migration [25–27]. The spatial organization of these cells—such as the proximity of HSCs to sinusoids, megakaryocytes, and bone surfaces—further influences their function and fate [28, 29]. The BMM adapts to physiological needs (e.g., injury, inflammation) by altering cellular composition and signaling, which can shift hematopoietic output or bone remodeling [25, 30]. Disruption of

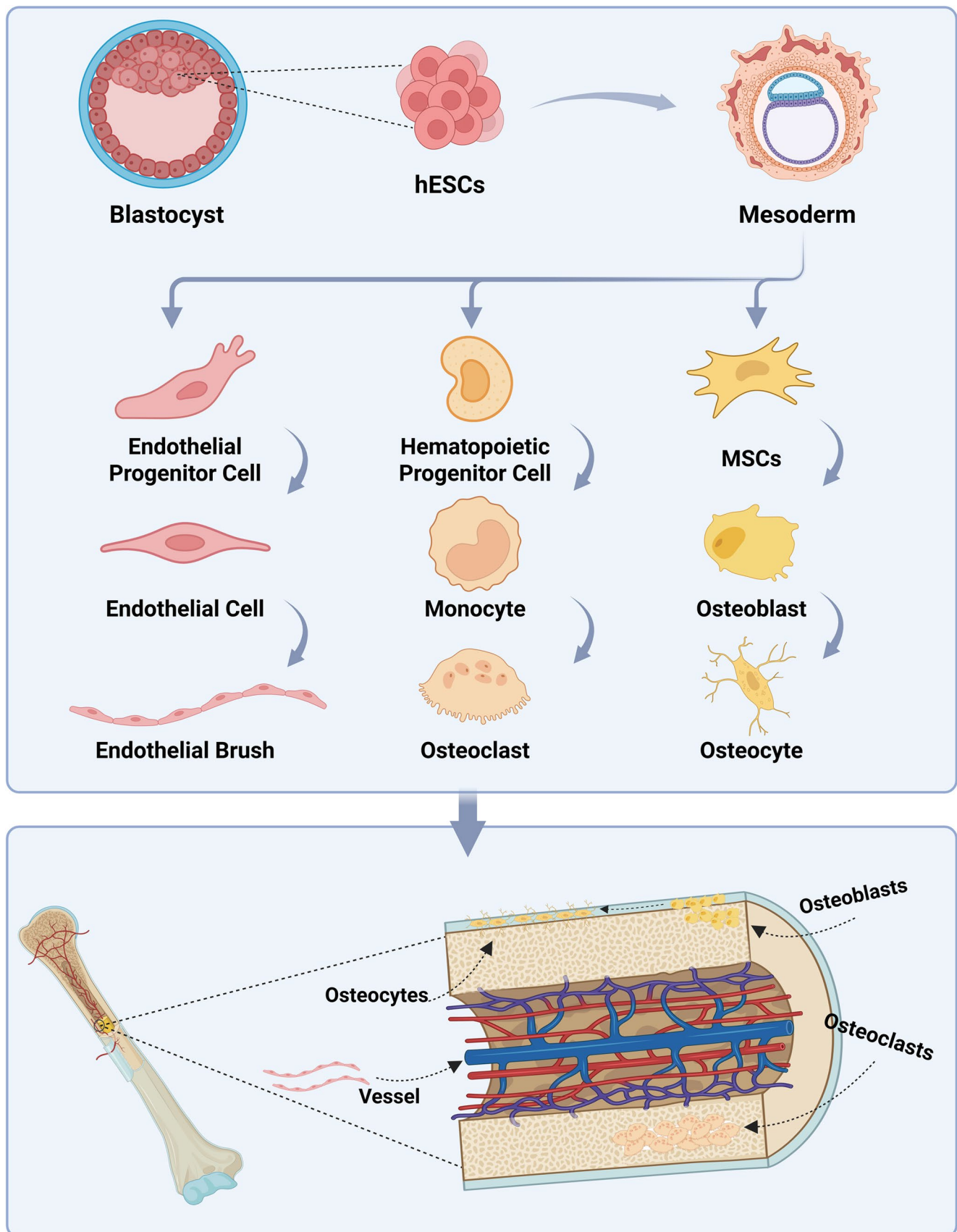


Fig. 2 Schematic overview of in vivo hESC-derived lineage specification and natural bone tissue architecture

this equilibrium—through aging, chronic inflammation, or malignancy—can lead to hematological diseases, impaired bone health, or altered immune responses [31, 32].

***In Vivo* Bone Development Process**

The mammalian skeleton contains both cartilage and bone. Both chondrocytes and osteoblasts are derived from hMSCs to form cartilage and bone tissues. Bone formation occurs via intramembranous osteogenesis and endochondral osteogenesis. Intramembranous osteogenesis is found in bones such as cranial bones, some clavicles and mandibles. Intramembranous osteogenesis begins with the polymerization of hMSCs, which differentiate into a vascular-rich hoof tissue membrane [33]. Then, they migrate to the osteogenesis site to differentiate into preosteoblasts in response to induced signaling molecules and further mature toward osteoblasts. Osteoblasts can form osteoids by secreting collagen fibers and an extracellular matrix by depositing inorganic salts such as calcium and phosphorus. Intracartilaginous osteogenesis is the primary mode of bone formation in certain irregular, long, and short bones. For example, the mesodermal mesenchymal cells of long bones accumulate at the osteogenesis site and differentiate into chondrocytes, followed by the formation of hyaline cartilage plates through the secretion of the extracellular matrix [34]. Since seed cells produced via bone tissue engineering have direct osteogenic ability, this review describes mainly the osteogenic method of intramembrane osteogenic differentiation *in vivo*.

***In Vivo* Derivation of Osteoblasts, Osteoclasts, and Vascular Endothelial Cells**

The key cellular components are derived from distinct developmental lineages: osteoblasts originate from hMSCs, osteoclasts from hematopoietic stem cells (HSCs), and vascular endothelial cells from endothelial progenitor cells. Through stepwise differentiation, hESCs progress through mesodermal intermediates into the above lineage-committed progenitors, hMSCs, HSCs, and endothelial progenitor cells, ultimately generating mature functional bone cell populations.

Osteoblasts, the primary cells deposited in the bone matrix, are derived from hMSCs originating from several different embryonic lineages and germ layers. Specifically, hMSCs that form the axial bone are derived from the paraxial mesoderm. The long bones of the limbs originate from lateral plate mesoderm cells, whereas craniofacial bones are derived from neural crest cells of the neural ectoderm [35]. These cells migrate to future skeletal sites and undergo bone formation through intramembranous ossification or

endochondral ossification. Notably, this differentiation process consists of several key stages: starting with hESCs, passing through the mesoderm and hMSCs into osteogenic progenitor cells, followed by the differentiation of osteogenic precursor cells, osteoblasts, and, ultimately, the formation of osteocytes.

Osteoclasts, the multinucleated cells responsible for bone resorption, are scattered on the surface of bone tissue and are strongly capable of undergoing osteolysis, phagocytosis, and digestion. Osteoclasts originate from the monocyte/macrophage lineage of HSCs and progress through the common myeloid progenitor (CMP) stage to become monocyte precursors. These precursors then migrate to the bone surface under the guidance of microenvironmental signals (e.g., cytokines and chemokines) and fuse to form mature multinucleated osteoclasts. Mature osteoclasts adhere to the bone matrix and secrete hydrogen ions and proteolytic enzymes, such as cathepsin K, to degrade mineralized bone, releasing calcium and phosphate into the circulation. In bone tissue, osteoblasts and osteoclasts complement each other and participate in bone growth and remodeling [36].

Bone is a highly vascularized tissue [23]. During embryogenesis, endothelial progenitor cells (EPCs) derived from the hemogenic endothelium of the mesoderm differentiate into mature endothelial cells under the regulation of the Notch and Wnt/ β -catenin signaling pathways [37]. In adult bone, functionally specialized vascular subtypes emerge: H-type vessels are predominantly localized in the metaphysis and periosteum, where they highly express proangiogenic factors such as vascular endothelial growth factor (VEGF) and platelet-derived growth factor-BB (PDGF-BB) [38]. These vessels colocalize with osteoprogenitor cells (Osterix⁺) to establish a “vascular-osteogenic niche”, directly promoting osteogenesis. In contrast, L-type vessels reside within the bone marrow cavity and are associated with adipogenic microenvironments, potentially regulating bone metabolism indirectly via paracrine bone morphogenetic protein 2 (BMP2) secretion [39].

The Construction of Bone Organoids Using Cells from hPSCs

Bone organs are microbone tissues with bionic characteristics constructed from the cellular components of natural bone tissue combined with related bioactive materials such as the extracellular matrix (ECM), hydrogels, and polymers. An analysis of the *in vivo* processes of bone formation in the second part revealed that bone formation, bone resorption, and hematopoiesis are the three main dynamic processes of bone regeneration. Accordingly, it is essential to obtain the following bone-related cell types: osteoblasts for

bone tissue construction, osteoclasts for bone remodeling, and vascular endothelial cells for nutrient and oxygen delivery. Different cytokines regulate the biological processes of osteogenesis in various ways. The appropriate addition of relevant growth factors helps the maturation and development of bone organoids.

The successful construction of bone organoids, in turn, can sequentially release growth factors that aid in repair. Recent research has demonstrated that bone organoids, as 3D culture systems, can secrete a variety of bioactive molecules that play a crucial role in promoting osteogenesis. For instance, BMPs and VEGF are known to induce osteogenic differentiation of hMSCs and osteoblasts, and have been found to be secreted by organoids, stimulating the migration of osteogenic precursors and blood vessels to fracture sites [40]. A study by Lin showed that organoids derived from hMSCs secreted BMP-2, which significantly enhanced osteogenic differentiation [41]. Similarly, human-derived bone marrow organoids successfully simulated key features of the bone marrow microenvironment by hiPSCs to differentiate into a 3D structure containing mesenchymal, endothelial, and hematopoietic cells. This model can autonomously secrete hematopoietic support factors such as SCF, FLT3L, GM-CSF, M-CSF, various interleukins (IL-2,3,4,6,7,8,11), and chemokines (CCL2-4) in the absence of exogenous factors, confirming the functional hematopoietic niche activity of its stromal cells [7]. Furthermore, organoids can secrete other extracellular matrix proteins, such as collagen and fibronectin, which help create a supportive microenvironment for osteogenesis [42].

To establish corresponding bone microenvironments, biomaterials that support cell growth under three-dimensional conditions were introduced. However, owing to their limited sources, immunogenicity, and ethical issues, hMSCs and progenitor cells cannot provide sufficient organoid constituent cells. However, fortunately, breakthroughs in hPSC-directed differentiation technology have provided revolutionary solutions. The organoid constituent cells derived from hESCs and hiPSCs are obtained by simulating sequential physiological signals during *in vivo* development, and they are expanded and differentiated through a multistep process.

Since the development of organoid culture techniques, the brain, liver, lung, and other organoids have been created and successfully used in the disciplines mentioned above. However, progress in bone organoid culture and establishment has been relatively slow because bone contains various soft and hard components, such as blood vessels and the bone matrix. Establishing at least two extracellular environments with different stiffness levels simultaneously is difficult. Nevertheless, some bone-related organoids have also been investigated and discovered [43].

Obtaining hPSC-Derived Cells from hPSCs

Differentiation of hPSCs into Osteoblast-Like Cells

Osteoblasts are derived primarily through direct induction via the intermediate stage of human pluripotent stem cell-derived mesenchymal stem cells (hPSC-MSCs). *In vivo*, this differentiation process is tightly regulated by a complex network of signaling pathways, including the Notch, Hedgehog, Bone Morphogenetic Protein-Small Mothers Against Decapentaplegic (BMP-Smad), Wnt/ β -catenin, Mitogen-Activated Protein Kinase (MAPK), Fibroblast Growth Factor (FGF), Ras Homolog Family Member A/Rho-associated Coiled-Coil Containing Protein Kinase (RhoA/ROCK), and Transforming Growth Factor- β (TGF- β) pathways [16, 44]. These pathways interact to modulate the osteogenic differentiation and bone regeneration/reconstruction functions of bone marrow-derived mesenchymal stem cells (BMSCs) (Fig. 3). *In vitro*, current induction strategies for osteoblast differentiation are generally categorized into two main approaches: embryoid body (EB) induction methods and monolayer induction methods. These techniques aim to mimic the *in vivo* signaling environment to promote osteogenesis effectively.

The EB induction method uses 3D cell aggregates to mimic the microenvironment of embryonic development, thereby supporting spontaneous multilineage differentiation [45]. During EB formation, hPSCs undergo mesoderm/ectoderm differentiation and generate hMSCs, subsequently achieving osteogenic induction under the action of three factors of osteogenesis, Dex (dexamethasone), β -glycerophosphate, and ascorbic acid (AA) [46]. The EB induction method presents a dual profile with advantages and limitations. The technique suffers from low differentiation efficiency because of the use of nonstandardized protocols and labor-intensive procedures. However, these 3D aggregates (EBs) from hPSCs precisely mimic human developmental processes and cell–cell interactions. By bridging the gap between *in vitro* research and *in vivo* processes, EB-based systems are ideal for constructing biomimetic models, such as bone organoids, bridging basic research and clinical applications [47].

In contrast, the monolayer induction method uses a 2D culture system with osteogenic-inducing medium (OM) to directly drive the directed osteogenic differentiation of hPSCs on culture plates. We highlight the simplicity and accessibility of the monolayer method, making it suitable for streamlined differentiation needs [48]. OM consists of basal culture medium, fetal bovine serum (FBS), β -glycerophosphate, ascorbic acid, dexamethasone, L-glutamax, nonessential amino acids (NEAAs), and antibiotics.

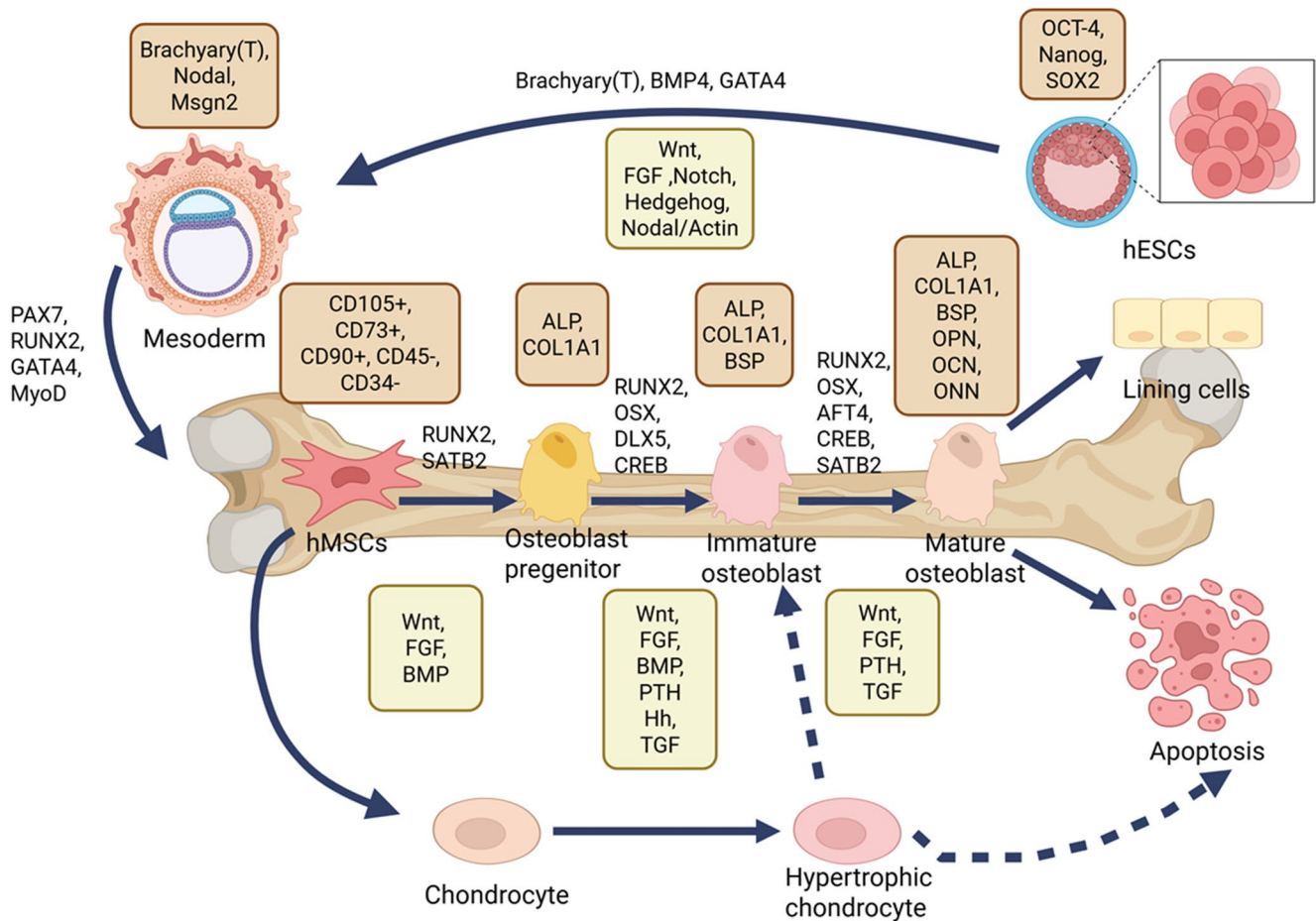


Fig. 3 The process by which hESCs differentiate into osteoblast-like cells. hMSC, human mesenchymal stem cell. hPSCs, human pluripotent stem cells. BMP, bone morphogenetic protein. RUNX2, runt-related transcription factor 2. FGF, fibroblast growth factor. PTH, parathyroid hormone. Hh, hedgehog. SATB2, special AT-rich sequence-binding protein 2. TGF, transforming growth factor. OSX, osterix. COL1A1,

collagen type 1 alpha 1 chain. ATF4, activating transcription factor 4. ALP, alkaline phosphatase. CREB, cAMP-responsive element-binding. OPN, osteopontin. ONN, osteonectin. OCN, osteocalcin. BSP, bone sialoprotein. Wnt, wingless/integrated. Pax7, paired box protein Pax-7. Notch, neurogenic locus notch homolog protein. GATA4, GATA binding protein 4. MyoD, myogenic differentiation

Studies have demonstrated that adding biological factors can accelerate the direct osteogenic induction of hPSCs. For example, in 2019, retinoic acid (RA) in the OM was shown to enhance the direct differentiation of hPSCs into osteoblast-like cells, potentially through stimulation of the bone morphogenetic protein (BMP) and WNT signaling pathways [49]. Additionally, rapamycin promotes Runt-Related Transcription Factor 2 (RUNX2) and BMP2 expression and activates BMP/Smad signaling through the inhibition of the AKT Serine/Mechanistic Target of Rapamycin/p70 Ribosomal Protein S6 Kinase (AKT/mTOR/p70S6K) pathway, even in the absence of osteogenic factors [50]. Furthermore, the synergistic role of biomaterials was confirmed: hPSCs cultured in strontium/cobalt-doped bioactive glass (BG)-containing conditioned medium exhibited significant upregulation of RUNX2, osteocalcin (OCN), and osteocalcin (ONN) expression [51].

The above traditional induction methods rely on undefined components such as Matrigel and FBS, which may induce immunogenicity and lead to inconsistent experimental outcomes. To address these issues, researchers have developed stepwise differentiation. In 2014, Ohba et al. noted that the undefined Matrigel surface used in their experiment could not maintain stable cell viability [52]. In 2020, they transferred hPSCs to vitronectin-coated surfaces for culture and successfully directed osteogenic differentiation by temporally sequencing the addition of distinct growth factors. This protocol minimizes confounding factors in the differentiation process, providing a more stable cell source for bone regeneration [53].

Notably, several studies have shown that hPSCs from different sources may contain specific epigenetic memories, thereby exhibiting some characteristics of high-priority differentiation. In 2017, hPSCs derived from osteoblasts exhibited a preferential differentiation tendency toward

osteogenic/chondrogenic lineages due to abnormal deoxyribonucleic acid (DNA) methylation, yet they lost the ability to generate adipocytes [17]. Selecting appropriate hPSC sources and adjusting culture conditions accordingly may increase differentiation stability.

Differentiation of hPSCs into Osteoclast-Like Cells

Osteoclasts, the primary cells responsible for bone resorption, originate from the monocyte–macrophage lineage [54–57]. The differentiation of these cells is tightly regulated

in vivo by two key signaling pathways: the macrophage colony-stimulating factor (M-CSF)/colony-stimulating factor 1 receptor (C-FMS) pathway and the receptor activator of nuclear factor- κ B ligand (RANKL)/receptor activator of nuclear factor- κ B (RANK) pathway. These pathways coordinate to promote the survival, proliferation, and differentiation of osteoclast precursors. In response to micro-environmental signals such as cytokines and chemokines, these precursors migrate to the bone surface and fuse to form mature, multinucleated osteoclasts capable of resorbing bone tissue (Fig. 4).

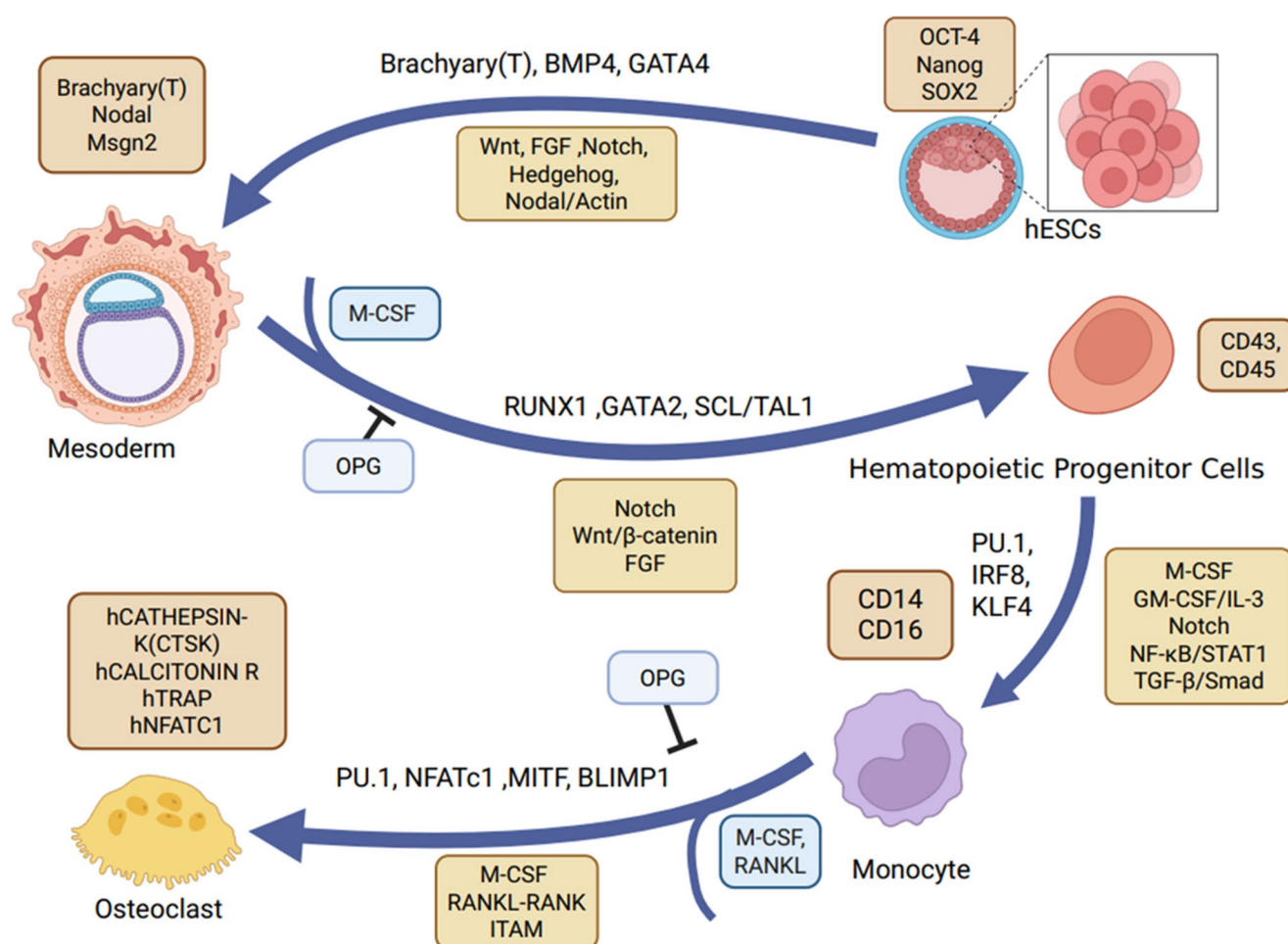


Fig. 4 The process of the differentiation of hESCs into osteoclasts. Brachyury (T): Brachyury (T-box transcription factor). Nodal: Nodal (TGF- β superfamily signaling molecule). Msgn2: Mesogenin 2. BMP4: Bone morphogenetic protein 4. GATA4: GATA Binding Protein 4. Wnt: Wingless-type MMTV integration site family. FGF: Fibroblast growth factor. Notch: Notch transmembrane receptor. Hedgehog: Hedgehog family. SCL/TAL1: Stem cell leukemia/T-cell acute lymphoblastic leukemia 1. RUNX1: Runt-related transcription factor (1) GATA2: GATA Binding Protein (2) IRF8: Interferon regulatory factor 8. KLF4: Krüppel-like factor 4. MITF: Microphthalmia-associated transcription factor. BLIMP1: B lymphocyte-induced maturation protein 1 (PRDM1). NFATc1: Nuclear factor of activated T cells, cytoplasmic 1. STAT1: signal transducer and activator of trans-

cription (1) Smad: Mothers against decapentaplegic homologs. OCT-4: Octamer-binding transcription factor 4 (POU5F1). Nanog: Nanog homeobox (pluripotency transcription factor). SOX2: SRY-Box Transcription Factor (2) M-CSF: Macrophage colony-stimulating factor. CD43: Leukosialin (cell surface glycoprotein). CD45: protein tyrosine phosphatase, receptor type C. OPG: osteoprotegerin (TNFRSF11B). RANKL: Receptor activator of nuclear factor kappa-B ligand. RANK: Receptor activator of nuclear factor kappa-B. ITAM: Immunoreceptor tyrosine-based activation motif. GM-CSF: granulocyte–macrophage colony–stimulating factor. IL-3: Interleukin-3. TGF-B: Transforming growth factor beta. CD16: Fc γ receptor III. PC3: Prostate cancer cell line (abbreviated as context-dependent)

The standard protocols for inducing osteoclasts *in vitro*—relying on primary precursors from bone marrow or peripheral blood with M-CSF and RANKL—face significant hurdles in efficiency and practicality [58]. When using primary human precursors (from bone marrow or peripheral blood) with M-CSF and RANKL, several inherent limitations arise. Firstly, these precursors exhibit considerable donor-dependent variability, leading to inconsistent differentiation outcomes between batches. Secondly, the resulting osteoclasts are terminally differentiated, preventing their expansion and limiting their use to a single experiment [59]. Moreover, as a genetic virtually untransfectable cell lineage, primary cell-derived osteoclasts and their precursors are poorly suited for genetic manipulation and promoter studies [60]. The RAW 264.7 murine cell line has been extensively utilized as a convenient *in vitro* model for studying osteoclast differentiation and function, due to its robust and reliable response to RANKL stimulation [61]. However, the model demonstrates functional limits compared to human osteoclasts, notably limited capacity of bone matrix acidification and collagen degradation in the case of bone perforation. Furthermore, the cell line's inherent heterogeneity and tendency for genetic drift can lead to substantial variability, often resulting in poor reproducibility across different laboratories and experimental batches [62]. hiPSCs present a better alternative source for OCs. As they can be propagated indefinitely, they allow for expandability and upscaling of OC production. Moreover, they have uniform initial state. This allows for the differentiation of large numbers of consistent genetic background and high synchrony of differentiation OCs, which minimizes intercellular variability and facilitates OC research.

In vitro, the differentiation of hPSCs into osteoclasts typically mimics this developmental progression, with hPSCs passing through the mesodermal stage, hematopoietic progenitors, monocytes/macrophages, and finally mature osteoclasts [63]. Among the reported induction strategies, the EB method remains the most commonly used approach for generating osteoclasts from hPSCs [64, 65].

In the direct differentiation process, as early as 2005, researchers reported that the initial addition of adenosine, cytidine, guanosine, uridine, deoxyadenosine, deoxyguanosine, thymine, deoxycytidine, sodium pyruvate, vitamin B12, lipoic acid, and biotin A.A. to hESCs and the subsequent addition of M-CSF and RANKL could result in osteoclast-like cells. Total cell recovery and osteoclast precursors were increased by increasing the number of fetal liver kinase-1 (FLK-1)-positive cells when ascorbic acid was added within the first 4 days [66]. The differentiation of hPSCs toward mesoderm has been extensively studied and is discussed in detail in the third part of this dissertation. The development of various differentiation protocols

has significantly increased the efficiency of hPSC differentiation toward the mesoderm. In the process of differentiating mesoderm into monocytes, researchers commonly refer to hematopoietic differentiation protocols. In this study, human stem cell factor (hSCF), human FMS-like tyrosine kinase 3 ligand (hFLT3L), human granulocyte colony-stimulating factor (hGCSF), human bone morphogenetic protein 4 (hBMP4), human interleukin-3 (hIL3), and human interleukin-6 (hIL6) were used [67]. *In vitro* erythrocyte expansion commonly employs hSCF and hIL3, with their optimal concentrations and addition times established in 2010. This process primarily involves the activation of cell surface receptors such as stem cell growth factor receptor (SCFR)/c-Kit and interleukin-3 receptor (IL-3R). Activation subsequently triggers multiple downstream signaling pathways, including the Janus kinase/signal transducer and activator of transcription (JAK/STAT), MAPK/extracellular signal-regulated kinase (ERK), and phosphatidylinositol 3-kinase (PI3K)/Akt pathways, among others [68]. Activating these signaling pathways facilitates cell proliferation, differentiation, and survival [59].

Notably, monocyte differentiation is also achieved by reducing the levels of growth factors such as L-glutamine, NEAAs, β -mercaptoethanol, hIL-3, and M-CSF. The process by which hPSCs differentiate into osteoclasts can be likened to the hematopoietic differentiation process [69]. In particular, signaling pathways involving FGF2, VEGF, RA, and TGF β have been demonstrated to regulate the transition from hESC-derived CD31 + endothelial cells to CD43 + hematopoietic cells [70, 71]. During the differentiation of monocytes into osteoclasts, the present study employed M-CSF and RANKL [72]. Although the time of differentiation and the concentration of growth factors varied slightly, the overall processes were similar across published studies [59, 65, 71, 73]. Osteoclastogenesis can be induced rapidly within 3–5 days by adding M-CSF and soluble receptor activator of nuclear factor- κ B (sRANK) [74]. A preprint study revealed no significant difference in osteoclast differentiation efficiency between peripheral blood mononuclear cells (PBMCs) and fibroblast-derived hiPSC lines [69]. Currently, the majority of media for osteoclast culture contain animal-origin FBS with unclear composition [75]. Few studies have developed chemically defined media. The use of vitamin D, low-density lipoprotein (LDL), and alternative serum can effectively promote osteoclast differentiation [76]. LDL can induce cell fusion and regulate the formation and survival of osteoclasts [77]. Vitamin D can increase the expression of the osteoclast adhesion molecules Integrin Alpha-V Beta-3 (α V β 3) and RANK [78]. However, the bone resorption ability of differentiated osteoclasts remains unsatisfactory, indicating a need for further improvement.

Recent studies have underscored the significant role of osteoclasts in developing organoids. The fracture healing process is intricate and involves various coordinated steps [79]. Various physiological processes, such as chondrocyte differentiation and the migration of osteogenic precursors and blood vessels to the fracture site, which are induced by immune responses from inflammatory cells, occur following fracture. Concurrently, osteoclasts infiltrate soft healing tissue and initiate the breakdown of the mineralized cartilage matrix. The delicate balance between matrix resorption and bone tissue formation results in the substitution of cartilage scabs with osteosclerotic scabs, leading to a state of dynamic equilibrium [80]. Researchers frequently incorporate bone marrow macrophage (BMM)-derived osteoclasts into organoid constructions to increase resorption in models and to represent the overall equilibrium homeostatic properties of bone remodeling [72, 81]. In 2021, Lordachescu et al. successfully generated trabecular bone organoids by introducing primary osteoblasts and osteoclasts into the femurs of cattle [22]. In 2023, researchers were able to achieve this goal by constructing trabecular-like organoids containing primary osteoblasts and osteoclasts that can secrete sclerostin, which is regulated by FGF23 and the parathyroid hormone (1–34) fragment (PTH1-34) [82]. These trabecular organoids offer novel perspectives for investigating pathological bone loss and understanding the intricate and dynamic regulation of bone remodeling.

Differentiation of hPSCs into Endothelium-Like Cells

In vivo, the differentiation and function of endothelial cells are regulated by multiple signaling pathways, including the VEGF/VEGFR, Notch, BMP, and Wnt pathways [83, 84]. Figure 5 provides a schematic overview of this in vivo developmental process, outlining the key signaling cascades and stage-specific markers that define the transition from hESCs to mature endothelial cells. The rationale for this figure is to illustrate the biological blueprint that current in vitro differentiation strategies aim to recapitulate. Generating a stable vascular network from hPSC-derived endothelial cells is a critical prerequisite for engineering functional bone organoids, as this network is essential for nutrient transport, oxygenation, and supporting osteogenesis [85]. These signaling cascades coordinate to guide vascular development, endothelial specification, and the formation of functional vasculature, which are critical for supporting bone formation and regeneration within bone organoids. hPSCs serve as promising cellular sources for generating endothelial-like cells in vitro. Current methods for deriving EC-like cells from hPSCs include coculture induction, EB-based differentiation, and monolayer-based approaches [86]. Among these methods, the coculture and EB methods face limitations for

clinical translation because of their low induction efficiency, complex culture conditions, and requirement for mechanical dissociation to obtain single-cell suspensions. In contrast, monolayer-based methods offer a more scalable and defined system, making them more suitable for downstream applications such as tissue engineering and regenerative medicine.

Monolayer-based protocols have become the prevailing approach for producing endothelial cells from hPSCs, offering benefits such as ease of operation, relatively high yield, and better control of the development of an endothelial-like cell phenotype [87]. In 2014, Lian et al. reported that the application of CHIR-99,021 (CHIR) to hPSCs can induce the formation of brachyury + mesodermal cells, which are induced to differentiate into EC-like cells in endothelial growth media (VEGF-A and bFGF) [88]. In 2015, Patsch et al. demonstrated that adding BMP-4 or forskolin to CHIR is effective [89]. To achieve cell purification, Neuropilin-1 cells were sorted from cardiovascular progenitors, which also promoted the formation of EC-like cells [90]. In 2017, it became possible to specify mesodermal precursors to produce specific EC-like cells by modulating BMP-4 and activin A levels [87]. However, induced endothelial cells quickly undergo dedifferentiation and lose several of their distinctive properties within a few passages in culture, which is a significant deficiency that should be solved [89].

The incorporation of the above-described obtained EC-like cells into organoids for subsequent in vitro organoid vascularization occurred through coculturing with constituent cells or by adding EC-like cells after establishing organoids to enhance the vascular network and functionality of the bone organoid tissue [91].

Purification of hPSC-Derived Cells

The differentiation of human pluripotent stem cells (hPSCs) involves multiple steps and intricate pathways. Without undergoing cell purification, this process may lead to reduced differentiation efficiency, elevated tumorigenic risk, and compromised clinical applicability [92]. Thus, cell purification serves as a critical step to ensure safety and efficacy in downstream applications.

Conventional cell purification methods, such as enzyme digestion, differential adhesion and single-cell cloning, take advantage of differences in cell tolerance to digestive enzymes [93]. Therefore, when cultured cells are digested, fibroblasts fall off the wall first, while osteoblasts take a longer time to digest. This difference can be used to purify osteoblasts via multiple differential digestion methods. Compared with that of osteoblasts, the attachment process of fibroblasts is faster, and most of the cells often complete the attachment process in 10–30 min. Thus, the purification

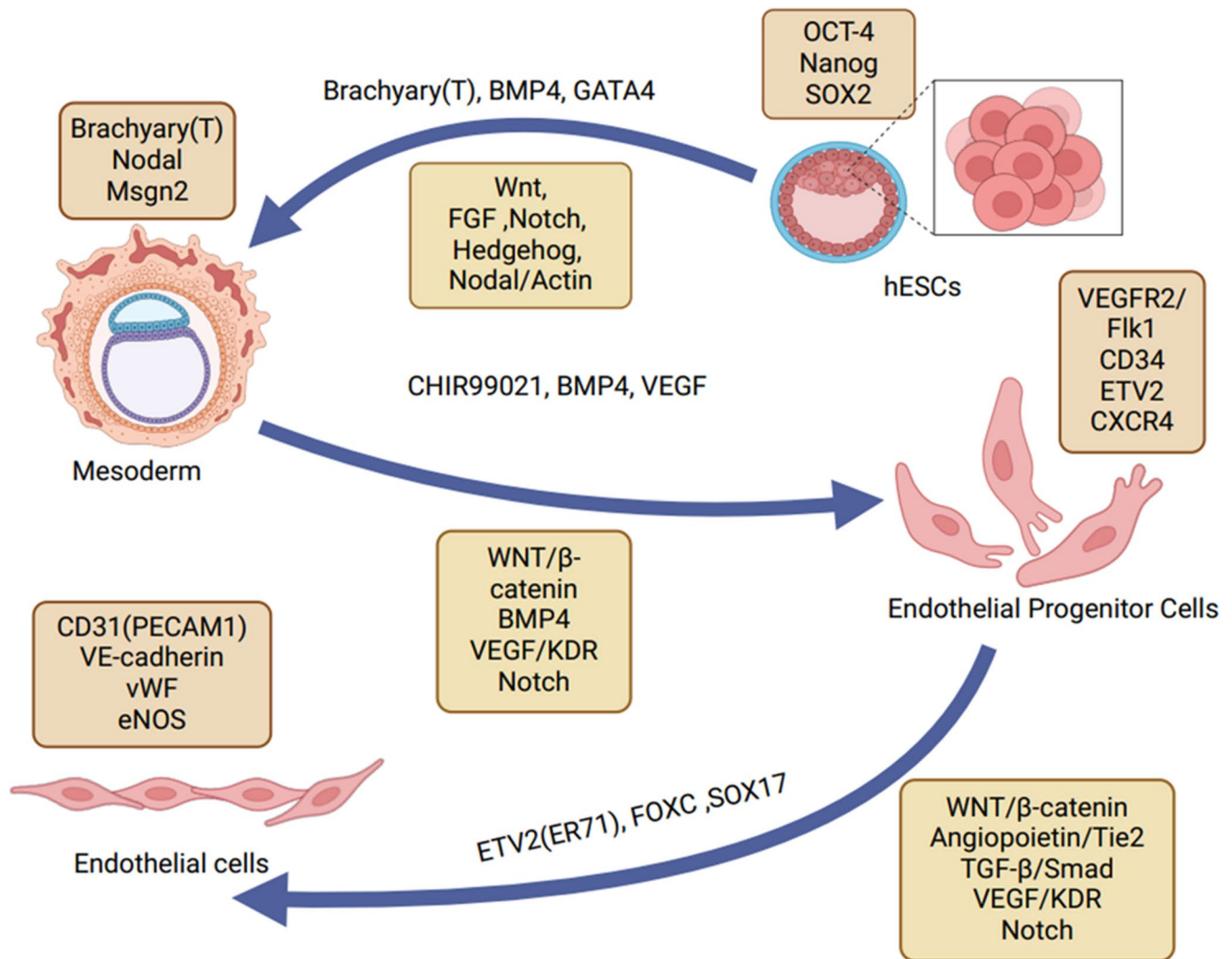


Fig. 5 The process by which hESCs differentiate into vascular endothelial cells. Brachyury (T): Brachyury (gene). BMP4: Bone morphogenetic protein 4. GATA4: GATA Binding Protein 4. Msgn2: Mesogenin 2. FGF: Fibroblast growth factor. VEGF: Vascular Endothelial Growth Factor. OCT-4: Octamer-Binding Transcription Factor 4 (POU5F1). Nanog: Nanog homeobox. SOX2: SRY-Box Transcription Factor 2. VEGFR2/Fik1: Vascular Endothelial Growth Factor Receptor 2/Kinase Insert Domain Receptor. CD34: Cluster of differentiation 34 (surface glycoprotein). ETV2: ETS Variant Transcription Factor 2.

CXCR4: C-X-C chemokine receptor type 4. CD31 (PECAM1): Platelet endothelial cell adhesion molecule 1. VE-cadherin: Vascular endothelial cadherin (CDH5). vWF: von Willebrand factor. eNOS: Endothelial nitric oxide synthase (NOS3). WNT/β-catenin: WNT/β-catenin (signaling pathway). VEGF/KDR: Vascular Endothelial Growth Factor/Kinase Insert Domain Receptor. FOXC: Forkhead box C1/C2 (transcription factor family). SOX17: SRY-Box Transcription Factor 17. Angiopoietin/Tie2: Angiopoietin (ligand)/Tyrosine Kinase with Immunoglobulin-like and EGF-like Domains 2 (receptor)

of osteoblasts can be achieved through repeated attachment [94]. However, they can only maximize the inhibition of the growth of cells other than osteoblasts but fail to accurately locate and identify the osteoblast group.

Therefore, the use of specific markers of osteoblasts for accurate screening is highly needed to obtain hPSC-derived bone organoid constituent cells [95]. The currently commonly used techniques include fluorescence-activated cell sorting (FACS), magnetic-activated cell sorting (MACS) and microfluidic cell sorting technologies (MST), which are the primary cell-sorting techniques, each of which requires live-cell analysis to preserve progenitor properties during

isolation [96]. Studies have shown that osteoblasts at different differentiation stages express distinct markers. Early precursor cells highly express RUNX2 and alkaline phosphatase (ALP), while the STRO-1 protein can serve as a potential marker [53, 95, 97]. Gronthos et al. demonstrated that cells expressing only the STRO-1 antigen (STRO-1/ALP-) exhibited functional characteristics of preosteoblasts, as their capacity to generate mineralized bone matrix diminished with the passage of time, and the combination of STRO-1/ALP represents the mature osteoblast stage [98]. Finally, mature osteoblast-like cells presented the highest expression levels of alkaline phosphatase, tissue-nonspecific

(ALPL), collagen type I alpha 1 chain (COL1A1), osteocalcin and interferon-induced transmembrane protein 5 (IFITM5) [99–101].

Like those used for osteoblast purification, specific markers are indispensable for the precise isolation of osteoclast and endothelial lineages. Nicholson et al. utilized CD14⁺ surface marker-based MACS to efficiently isolate monocytes from PBMCs and confirmed that CD14⁺ monocytes are pivotal preosteoclasts [102]. For mature osteoclasts, tartrate-resistant acid phosphatase (TRAP), OC-associated receptor (OSCAR), $\alpha\beta$ 3 and calcitonin receptor (CTR) are considered to be specific markers [103–105]. However, the specificity of these markers remains debated: TRAP is also expressed in macrophages and dendritic cells, the OSCAR may overlap with immune lineages, and $\alpha\beta$ 3 integrin is shared by endothelial and tumor cells, necessitating multiple markers and functional validation in osteoclast purification. Liu et al. developed a two-step MACS protocol using both CTR and OSCAR to efficiently isolate mature osteoclasts (OSCAR⁺CTR⁺) directly from mouse bone marrow [104].

Endothelial progenitors (EPCs) are typically enriched via CD34⁺/VEGFR2⁺ sorting. On this basis, mature endothelial cells are characterized by CD31 (PECAM-1)/CD144 (VE-cadherin) coexpression, whereas CD34-negative cells do not express endothelial properties [106, 107]. Langer et al. obtained vascular progenitor cells induced from hPSCs through MACS-based CD34⁺ selection, which were differentiated into endothelial or smooth muscle-like cells, and isolated mature endothelial cells via FACS sorting of CD31⁺ cells [108]. Although CD31⁺ is a core marker of mature endothelial cells, it overlaps with hematopoietic cells such as platelets and needs to be excluded by multiparameter sorting (e.g., CD45-/CD31+/CD144+) [109]. In addition, von Willebrand factor (vWF) is a specific marker of mature endothelial cells [110]. (Table 1)

Signaling Pathways during the Development of Bone Organoids

Although the two-dimensional differentiation protocols mentioned above provide a reliable means to generate the required cellular components—osteoblasts, osteoclasts, and vascular endothelial cells—from hiPSCs, they inherently lack the complex three-dimensional architecture

and continuous cell-cell and cell-matrix signaling in natural bone tissue. The promise of bone organoid technology contribute its ability to construct a sophisticated 3D micro-environment. When these individually derived cells are co-cultured within a biomimetic ECM scaffold, or when hiPSCs are guided through directed differentiation within a 3D matrix, they undergo a process of self-organization and functional integration. This transition from a collection of isolated cells to an integrated osteoid tissue introduces a new technology, regulated by spatiotemporally signaling networks.

The subsequent section will focus on elaborating the key signaling pathways including Wnt/ β -catenin for initial cellular aggregation, BMP-SMAD and MAPK for osteogenic differentiation and maturation, RANKL/RANK/OPG for bone remodeling, and VEGF for vascularization. Their interplay eventually drives the development, maturation, and homeostatic function of bone organoids (Fig. 6).

The development and maturation of bone organoids are regulated by key signaling pathways in a precise spatiotemporal sequence. At the initiation stage, the Wnt/ β -catenin pathway stabilizes β -catenin, facilitating its nuclear translocation where it associates with TCF/LEF transcription factors to activate mesodermal and mesenchymal lineage genes, while inducing epithelial-mesenchymal transition to promote cellular aggregation [117]. The TGF- β /BMP pathway acts synergistically with Wnt signaling; specifically, the BMP-SMAD cascade (via the SMAD1/5/8–SMAD4 complex) directly activates the master osteogenic transcription factor RUNX2 and its downstream effector Osterix [118], thereby driving mesenchymal specification and early osteogenic differentiation. Non-canonical pathways (e.g. MAPK, PI3K/AKT, and RhoA/ROCK) further modulate cell proliferation, survival, and cytoskeletal reorganization [119]. Within the MAPK family, the ERK pathway—activated by growth factors—initially promotes the expansion of osteoprogenitors while transiently suppressing differentiation; at later stages, it enhances RUNX2 transcriptional activity through phosphorylation at specific residues (e.g. Ser301 and Ser319). In contrast, the p38 pathway, responsive to BMP and cellular stress signals, directly phosphorylates RUNX2, Osterix, and ATF4, advancing osteogenic gene expression and cellular differentiation [120, 121]. During the maturation phase, the RANKL/

Table 1 Surface markers related to the purification of bone constructed cells

Cell type	Marker			Sorting technology	Ref.
Preosteoblast	ALP	STRO-1	RUNX2	MACS, FACS	[97, 98, 111, 112]
Mature Osteoblast	COL1A1	Osteocalcin	IFITM5	FACS	[99, 101]
Preosteoclast	CD11b+	CD14 ⁺		MACS	[102, 113]
Mature Osteoclast	TRAP	OSCAR	CTR	MACS	[103, 104, 114]
Endothelial progenitors	CD34 ⁺	VEGFR2 ⁺		MACS	[107]
Endothelial Cell	CD31 ⁺	CD144 ⁺	vWF	MACS, FACS	[109, 110, 115, 116]

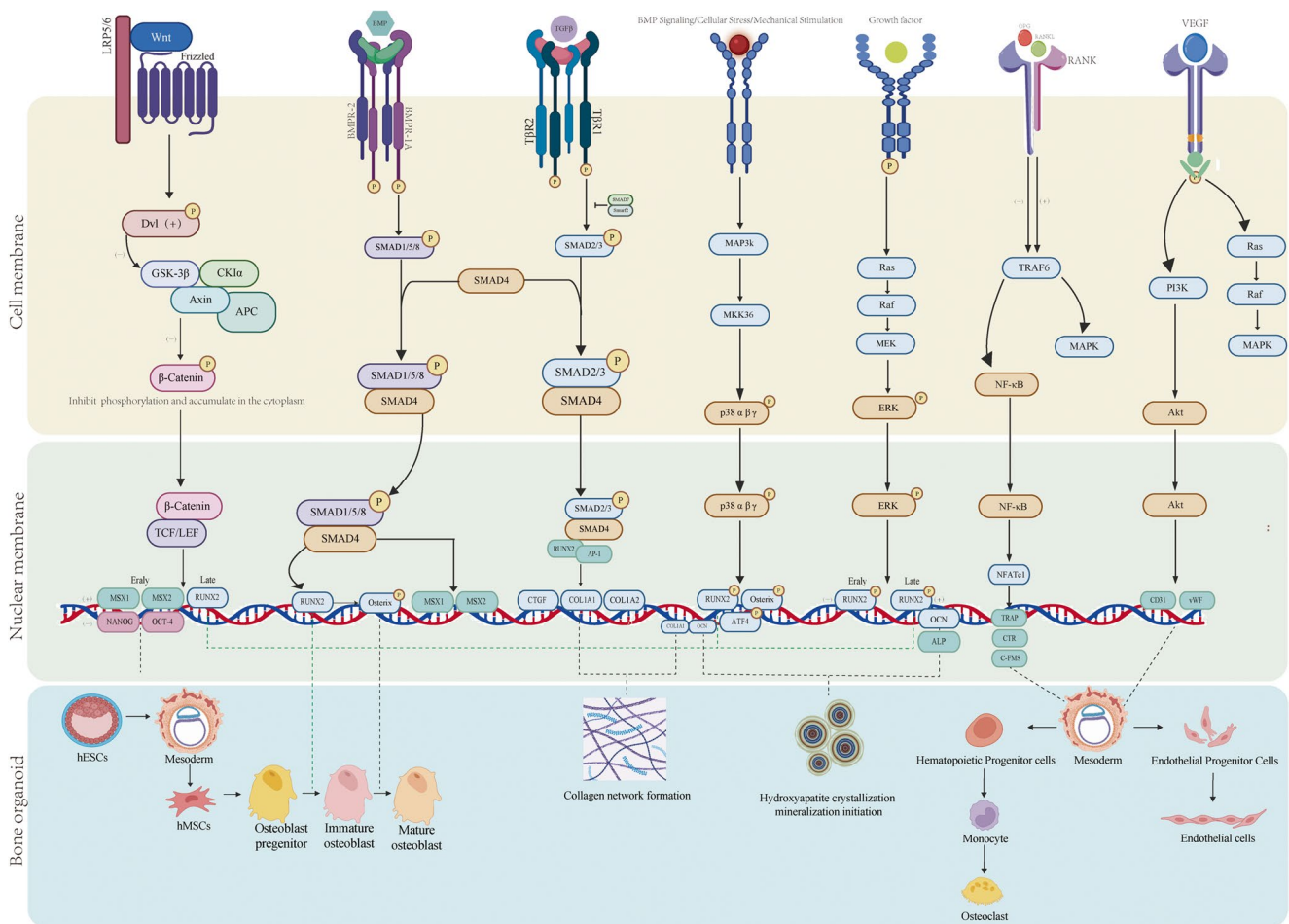


Fig. 6 Schematic overview of the key signaling pathways regulating the development and maturation of bone organoids. LRP5/6, low-density lipoprotein receptor-related protein 5/6. WNT, wingless-related integration site. FRIZZLED, frizzled receptor. DVL, dishevelled. GSK-3 β , glycogen synthase kinase 3 beta. CK1 α , casein kinase 1 alpha. AXIN, axin protein. APC, adenomatous polyposis coli. β -CATENIN, beta-catenin. TCF/LEF, t cell factor/lymphoid enhancer-binding factor. RSPO, r-spondin. R-SPONDIN, r-spondin protein. ZNF3, zinc-finger protein 3. RNF43, rnf43 ring finger protein. BMP, bone morphogenetic protein. TGF- β , transforming growth factor beta. SMAD1/5/8, small mothers against decapentaplegic 1/5/8. SMAD4, small mothers against decapentaplegic 4. SMAD2/3, small mothers against decapentaplegic 2/3. MAPK, mitogen-activated protein kinase. MKK36, mitogen-activated protein kinase 36. P38 $\alpha/\beta/\gamma$, p38 mitogen-activated protein kinase alpha/beta/gamma. MEK, mitogen-activated protein kinase. ERK, extracellular signal-regulated kinase. RAS, rat sarcoma. RAF, rapidly accelerated fibrosarcoma. GROWTH FACTOR, growth factor proteins. TRAF6, tnfr-associated factor 6. NF- κ B, nuclear fac-

tor kappa-light-chain-enhancer of activated b cells. VEGF, vascular endothelial growth factor. RANK, receptor activator of nuclear factor kappa- β . PI3K, phosphoinositide 3-kinase. MTOR, mechanistic target of rapamycin. NRPS6, nuclear ribosomal protein S6. TGF- β 1, transforming growth factor beta (1) TGF- β 2, transforming growth factor beta (2) TGF- β 3, transforming growth factor beta (3) TGF- β 4, transforming growth factor beta (4) CDK, cyclin-dependent kinase. PDGF-L1, platelet-derived growth factor receptor ligand 1. CYTOKININ, cytokinin. CTGF, connective tissue growth factor. COL1A1, collagen type i alpha 1 chain. COL3A1, collagen type iii alpha 1 chain. ALP, alkaline phosphatase. OCN, osteocalcin. OCS, osteocytes. ON, osteonectin. MSX1, msh homeobox. MSX2, msh homeobox. OCT4, octamer-binding transcription factor 4. TRAP, tartrate-resistant acid phosphatase. CTR, cation transporter. C-FMS, Colony-stimulating factor 1 receptor (CSF1R). CD31, cluster of differentiation 31. VDF, vascular development factor. BMP, bone morphogenetic protein. OPG, osteoprotegerin. RANKL, receptor activator of nuclear factor kappa- β ligand. SMURF2, smad ubiquitination regulatory factor 2

RANK/OPG axis precisely regulates bone remodeling: RANKL binding to RANK on osteoclast activates NF- κ B, MAPK, and NFATc1 pathways, inducing osteoclast differentiation and activation, whereas OPG serves as a decoy receptor that competitively inhibits this interaction [55]. The dynamic balance among these three components maintains the resorption and formation of bone. Simultaneously,

the VEGF pathway promotes angiogenesis via VEGFR2-mediated activation of ERK and PI3K/AKT signaling, supplying nutrients and oxygen to the organoid [122, 123]. Moreover, VEGF directly promotes osteoprogenitors differentiation and supports their survival, integrates vascularization with osteogenesis to ensure the structural and functional maturation of bone organoids.

Materials Suitable for the Fabrication of Bone Organoids

The two main types of 3D cell culture methods are scaffold-free systems and scaffold-based systems. Scaffold-free systems rely on certain cell types' ability to self-aggregate, which can be promoted by employing particular cell plates and physical conditions that prevent cell attachment. Instead, in scaffold-based systems, cells are planted in synthetic or natural materials, enabling cell aggregation, proliferation, and three-dimensional structure.

Scaffold-Free Methods

Scaffold-free methods for bone organoid generation rely on the innate self-organization and self-assembly capabilities of cells to form three-dimensional (3D) microtissues. The established techniques can be broadly categorized into static and dynamic systems, each with distinct operational principles and applications.

Among static methods, pellet culture represents the most straightforward technique, wherein dissociated cells are compacted via centrifugal force into a high-density aggregate at the bottom of a conical tube [124]. The hanging drop method offers superior control over spheroid size and uniformity. In this approach, small volumes of cell suspension (typically on the microliter scale) are pipetted onto the lid of a culture dish, which is then inverted. Gravity causes cells to settle at the liquid-air interface, promoting their aggregation into a single, well-defined spheroid per droplet [125]. The liquid overlay technique utilizes non-adherent surfaces (e.g., ultra-low attachment plates or AggreWell™ plates) to prevent cell attachment over a larger surface area [126]. This setup encourages cell-cell contact and ECM production as spheroids form under static or mildly agitated conditions.

Dynamic methods introduce physical forces to enhance the culture process. Magnetic levitation involves incubating cells with biocompatible magnetic nanoparticles, which associate with the cell membrane. When subjected to an external magnetic field, these labeled cells are levitated against gravity, facilitating rapid 3D aggregate formation in a scaffold-free manner, which promotes tissue-like compaction and signaling [127]. Bioreactor mediated cultivation represents a more advanced dynamic strategy. Systems such as rotating wall vessels (RWV) maintain cell aggregates in a state of continuous free-fall, creating simulated microgravity that minimizes shear stress while ensuring efficient nutrient/waste exchange. Similarly, spinner flasks and perfusion bioreactors provide controlled fluid dynamics and mechanical stimulation (e.g., fluid shear stress), which are critical for enhancing the viability, scalability, and osteogenic maturation of larger bone organoid constructs [128, 129].

The selection of a scaffold-free method depends on the specific research objectives, with considerations for spheroid uniformity, throughput, scalability, and the need for incorporating specific biophysical cues.

Scaffold-Based Methods

A multitude of scaffolds that can be utilized as templates for the production of tissues are available owing to advancements in regenerative medicine based on synthetic biomaterials. These scaffolds need to have certain properties, including chemical composition, shape, structure, and porosity, to facilitate cell migration, adhesion, and tissue creation.

Decellularized scaffolds composed of normal ECM components are the primary representations of natural scaffolds, and the main constituents of the ECM include collagen, elastin, laminin, fibronectin and glycosaminoglycan, which release a number of growth factors, cytokines and chemokines to regulate tissue development and homeostasis [130]. Therefore, it is necessary to simulate the ECM characteristics of tissues when developing 3D organoid models *in vitro*. Because the culture requirements of hPSC-derived cells *in vitro* are different from those of primary cells, the scaffolds used for bone organoids usually need modifications such as Matrigel, proteins or synthetic coatings. These biomaterials should not only support cell colonies rather than single cells but also regulate cell fate. For example, nanoscale surface topography (i.e., roughness, pattern and porosity) significantly affects cell shape, migration, proliferation and osteogenic differentiation [131]. Furthermore, relatively rigid bone organoids, as models simulating osteogenesis, mineralization and remodeling, are different from other soft tissue organoids, and certain mechanical properties are needed for bone defect repair. The adhesion, diffusion, migration and differentiation of stem cells are also affected by the mechanical properties of the matrix microenvironment. For example, hMSCs on hard substrates tend to differentiate into osteoblasts [132]. Moreover, bone tissue contains a complex vascular system, which is a typical layered endothelial cell network, and the development of biomaterials with vascular remodeling potential is still the top priority of bone organoids. Related strategies include the use of biomaterials with micro/nanostructures, different porosities, matrix stiffnesses, morphologies, and tubular structures and the controlled release of specific biological factors. It has received special attention because of its key impact on endothelial cell behavior and angiogenic potential (Fig. 7). These biomaterials should also provide a better culture environment so that the central transport of oxygen and nutrients is sufficient, thus expanding the growth of hPSC-derived organoids.

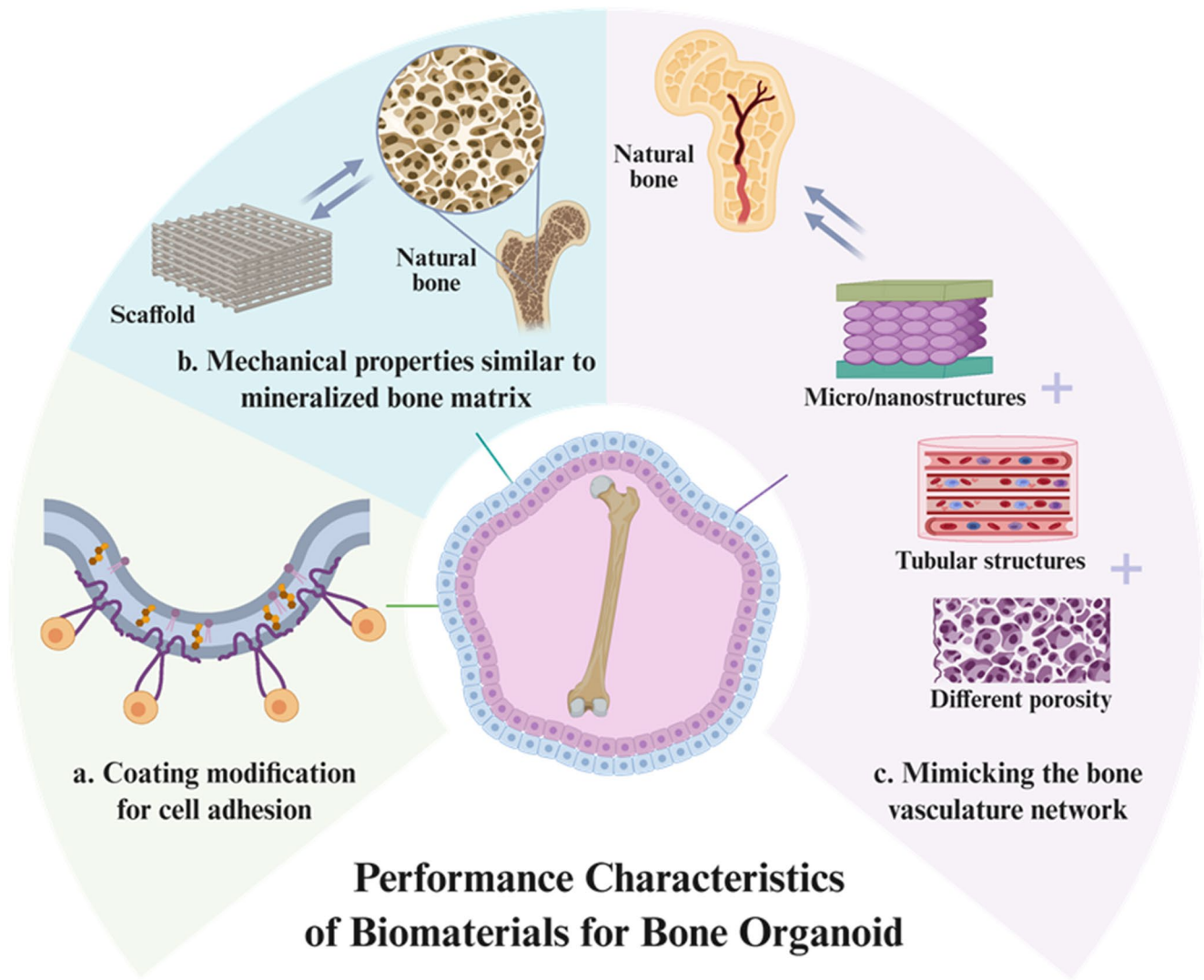


Fig. 7 The essential performance characteristics of biomaterials required for the construction of bone organoids, which can be categorized into three main aspects. **(a)** Surface coating modification enhances cell adhesion and promotes stable interactions between cells and scaffolds, facilitating cell proliferation, differentiation, and tissue integration. **(b)** Appropriate mechanical properties that approximate those of mineralized bone matrix are essential to provide structural

There are a variety of substitutes for ECMs, and the most commonly used substrate is Matrigel, which is a gelatinous protein mixture. It contains a mixture of ECM components, including laminin, type IV collagen, actin, heparan sulfate proteoglycans and some growth factors, such as TGF- β and FGF [133]. In 2021, Gu et al. proposed that Matrigel can simulate natural tissue and support the construction and growth of bone organoids [134]. However, Matrigel generated from animals is susceptible to degradation throughout bone organoid growth, resulting in insufficient mechanical support that falls short of the standard level [135]. Therefore, natural hydrogels with physical and chemical properties

stability and transmit mechanical cues regulating osteogenic differentiation. **(c)** Biomaterials designed with micro- and nanostructures, tubular arrangements, and varied porosity can replicate the hierarchical organization of native bone tissue. In particular, the presence of interconnected porous networks supports nutrient and oxygen exchange, assists angiogenesis, and enables the establishment of a functional vasculature system within the developing bone organoid

similar to those of ECM are ideal materials. Polysaccharides and proteins, including collagen, hyaluronic acid, gelatin, chitosan, sodium alginate, and heparin, are well recognized as the most common types of natural hydrogels [136]. For example, in 2019, Jee et al. created a matrix made of type 1 collagen and bicarbonate, which showed organoid culture abilities comparable to those of Matrigel in terms of cell proliferation and specific marker expression [137]. In 2020, a composite hydrogel capsule consisting of a core made of fibrin hydrogel and a shell made of an alginate-chitosan composite was created [138]; this technique facilitates the distinction and autonomous creation of liver organoids in an

uninterrupted process by enclosing hepatocyte-like cells obtained from hPSCs. In 2021, Luo et al. encapsulated colorectal cancer organoids in a well-defined 3D-hyaluronan-gelatin hydrogel [139]. However, their relative lack of stability and mechanical qualities should also be considered, while the utilization of cross-linking agents might enhance the mechanical characteristics of hydrogels, chemical cross-linking agents may pose a possible risk to nearby or enclosed cells. (Fig. 8)

Compared with biological materials, synthetic polymers are highly customizable and can be modified according to specific cultures by adding different proteins and

other molecules [140]. Synthetic hydrogels consist of hydrophilic polymers such as poly(ethylene glycol), poly(vinyl alcohol), poly(lactic acid), and polyacrylamide [135]. During synthesis, the binding site of matrix metalloproteinases can be modified to regulate the degradation rate of hydrogels. Similarly, the use of different preparation processes, such as electrospinning or microfluidics, has the ability to alter the morphology and dimensions of.

organoids [141]. In 2020, Vallmajo et al. combined polyethylene glycol and hyaluronic acid hydrogels, which had favorable physicochemical characteristics and notably enhanced the proliferation and specialization of

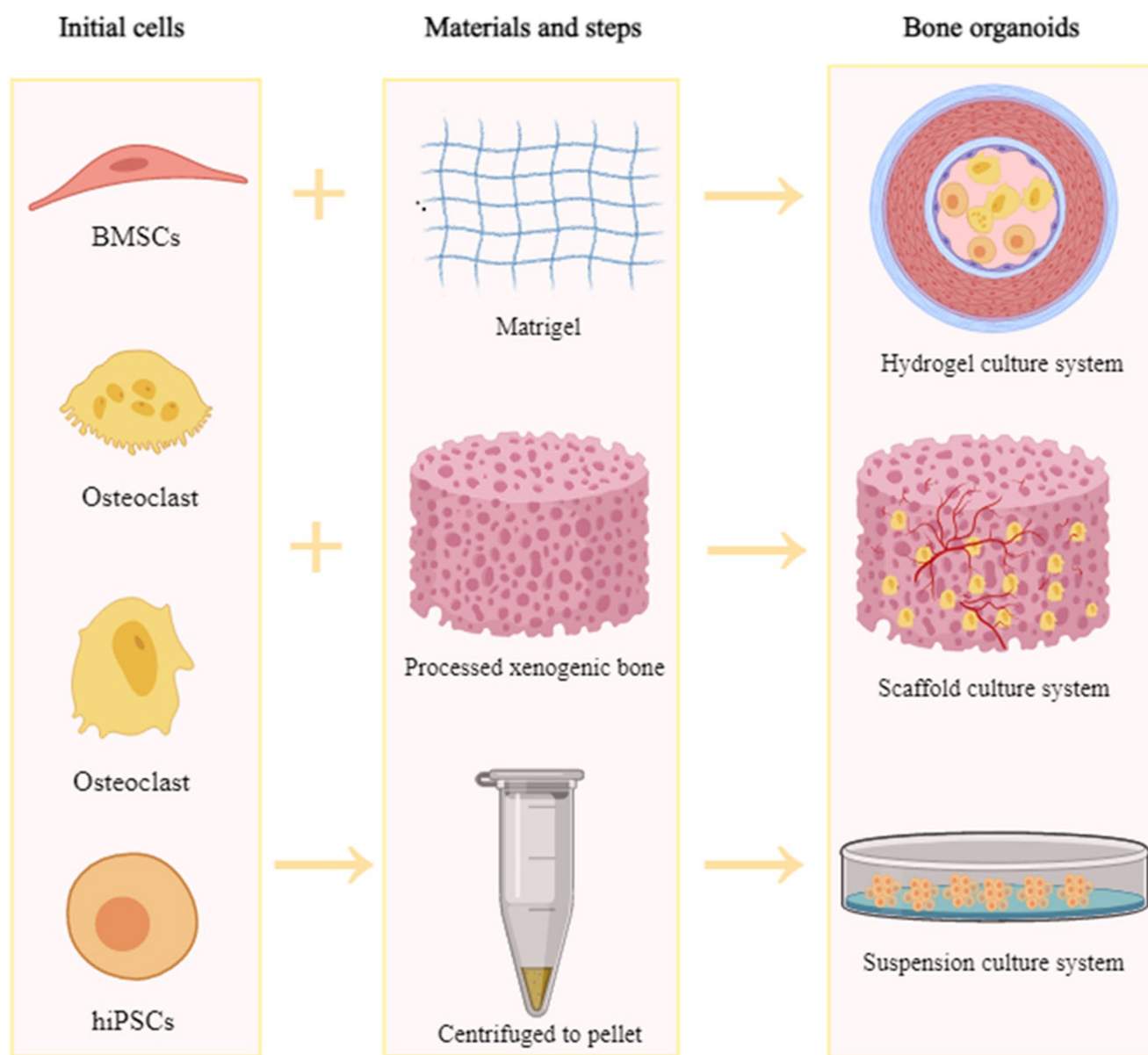


Fig. 8 Strategies for constructing bone organoids involve self-assembly within hydrogels, integration with processed xenogenic bone, and particle formation through centrifugation combined with suspension

culture, collectively enabling structural mimicry and functional maturation of bone-like tissues

BMSCs [142]. Dye et al. reported that lung organoids derived from hPSCs, when transplanted into scaffolds made of poly(lactide-co-glycolide) (PLG) or polycaprolactone (PCL), formed tube-like structures that closely resembled the structure and cellular diversity of an adult airway. This suggests that the interconnectedness of the pores in the scaffold and the degradation of the polymer play a role in the maturation process [143]. In 2022, in collaboration with bone marrow-derived stem cell (BMSC)-loaded hydrogel microspheres (MSs), Xie et al. created osteo-callus organoids for rapid bone regeneration via stepwise induction and digital light processing (DLP) printing technology [34]. As a new type of multi-tissue platform, the bone/cartilage organoid chip (BCoC) faithfully simulates biological functions and pathophysiological responses in real situations, which is expected to completely change the basic research and drug screening of diseases such as osteoporosis and arthritis [144]. (Fig. 9)

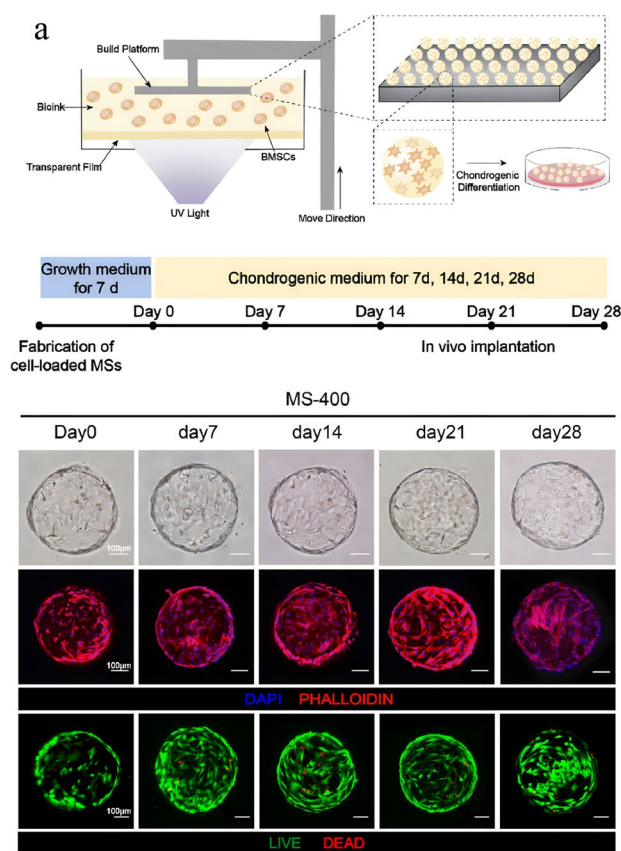
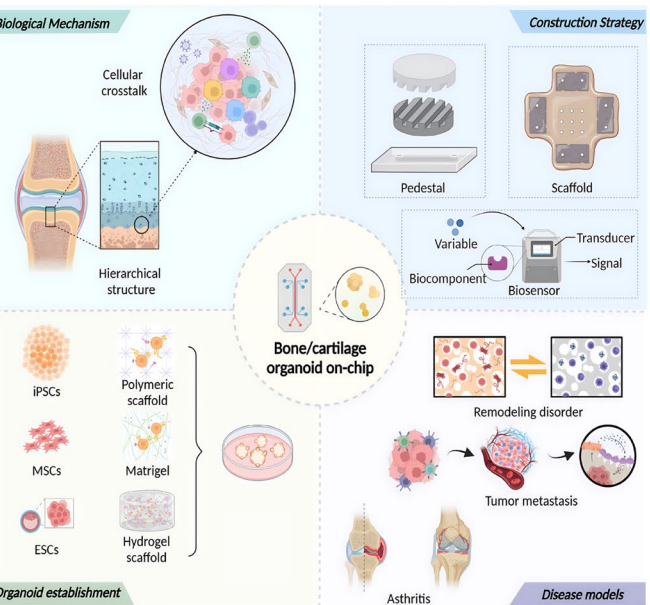


Fig. 9 Construction of bone organoids based on biomaterials: (a) Illustration of 3D printing of BMSC-loaded hydrogel microspheres (MSs) to produce osteo-callus organoids. Reproduced with permission [34]. Copyright 2022, Elsevier; (b) Fabrication of osteochondral (OC) or chondral (CH) tissue in the dual-flow bioreactor to imitate OA pathologies and test drugs. Reproduced under terms of the CC-BY license [145]. Copyright 2019, Lin, Li, Li, Li, Del Duke, Shen, Hao, O’

Towards a Practical Framework for Scaffold Comparison

While the preceding discussion highlights the vast landscape of scaffold materials, a critical challenge in the field lies in the evaluation and comparison of these diverse systems. The integration of various functional compounds, such as adhesion peptides and growth factors, creates a multi-parametric composites. This complexity precludes the application of a simple, uniform standardization. Instead, ensuring reproducibility and enabling meaningful cross-study comparisons requires a shift towards a multi-dimensional assessment framework.

A representative example of this challenge is the ongoing debate surrounding the “optimal” mechanical stiffness of hydrogels for osteogenic differentiation. While numerous studies indicate that a Young’s modulus in the range of 10 kPa to 100 kPa is generally conducive to osteogenesis, no universal “best” value exists [146]. The efficacy is highly context-dependent, influenced by factors such as whether



Donnell, Bunnell, Goodman, Alexander, Tuan and Lin, published by Frontiers; (c) Diagrammatic representation of the organoid on-chip for bone/cartilage. Reproduced under terms of the CC-BY license [144]. Copyright 2023, Yan Hu, Hao Zhang, Sicheng Wang, Liehu Cao, Fengjin Zhou, Yingying Jing, Jiacaan Su, published by KeAi Communications Co. Ltd

the hydrogel stiffness is static or dynamically tunable, incorporates specific biofunctional peptides (e.g., RGD), or is co-delivered with osteoinductive factors like BMP-2 [147–149]. The inconsistent reporting of key parameters—including measurement techniques, exact stiffness values, and concomitant biochemical cues—across different studies prevents meaningful comparison of results and hinders the identification of optimal combinatorial strategies.

In this context, the key is to move beyond cataloging a fixed set of material properties and toward justifying the selection of characterized parameters based on the scaffold's proposed mechanistic function. For instance, the reported value of hydrogel stiffness is functionally meaningless without complementary data on the measurement technique and the material's compositional background. Therefore, the reporting priorities should be context-driven: studies focusing on nutrient transport should prioritize quantifying pore interconnectivity and permeability, while research on mechanotransduction must provide biologically relevant stiffness metrics and ideally document their temporal evolution.

Adopting this nuanced perspective will provide a more rigorous and actionable foundation for the field. It emphasizes functional relevance over mere data collection, thereby facilitating a deeper understanding of scaffold performance and accelerating the development of more robust bone organoid models.

Advances in the Construction of Bone Organoids Using Cells from hPSCs

Organoids derived from hPSCs are often in an embryonic stage, which is more suitable for studying the state and physiology of early organs. At present, organoid technology derived from hPSCs has been applied in many disease fields. For example, hPSCs were effectively used to create kidney organoids for the purpose of studying human nephrogenesis and polycystic kidney disease (PKD) [150]. Cardiac organoids are utilized as a means to simulate the process of cardiomyocyte regeneration after injury [151]. Vascular organoids are generated from both hESCs and hPSCs and from vascular organoid models [152].

Although the technology for creating soft tissue organs is advancing, the development of bone organoids is still in its infancy. Compared with organs such as the intestines, stomach and brain, bone contains a variety of soft (bone marrow, blood vessels, etc.) and hard (mineralized bone matrix) components. Thus, the stiffness requirements of the cells are heterogeneous, and researchers may need to build at least two complex environments with different stiffness levels at the same time. The creation of bone organoids necessitates increased needs in various aspects, including specific medium composition, matrix, and mechanical stimulation.

In recent years, researchers have developed several bone organoid models, including bone marrow organoids, cartilage organoids, and bone trabecular organoids. The establishment of bone organoids via multidirectionally differentiated MSCs, ASCs or primary osteoblasts has been widely reported. For example, Scotti et al. generated bone organoids featuring a hematopoietic compartment with stem and progenitor cells, which they achieved using a BMSC-seeded collagen scaffold and IL-1 β treatment to guide development, resulting in a structure and function akin to natural bone [153]. In 2021, Iordachescu et al. created a prototype of a micron-sized bone organoid. This model combines primary female osteoblastic and osteoclastic cells, after which the structures are implanted into human tissue and subjected to simulated microgravity for 5 days. During this period, significant amounts of bone mass are lost from the initial structure, enabling further simulation of the bone loss process [22]. Su et al. utilized DLP-3D bioprinting technology with a bone matrix-inspired bioink to create a centimeter-scale bone organoid. This model integrated seven distinct bone marrow-associated cell types—including osteoblasts, osteoclasts, vascular endothelial cells, chondrocytes, adipocytes, immune cells, and blood cells—within a single self-mineralizing structure, thereby recapitulating a complex cellular microenvironment [154]. These organoid strategies provide a reference for the integration of osteogenic and vascular components.

On this basis, we specifically focused on bone organoids constructed with hPSC-derived cells. In 2019, Lin et al. used hPSCs to construct a biphasic osteochondral organoid in a dual-flow bioreactor. The bone compartment exhibited mineralization (Alizarin Red staining) and osteogenic activity (OCN expression), and the model demonstrated IL-1 β -induced remodeling with upregulation of MMPs and ADAMTS [145]. In 2021, O'Connor et al. generated scaffold-free osteochondral organoids from murine iPSCs through sequential differentiation using TGF- β 3 and BMP2 to mimic endochondral ossification. The resulting organoids demonstrated mineralization in the bony region (confirmed by micro-CT), expressed key osteogenic markers (Alpl, Bglap, Runx2), and exhibited tissue remodeling from cartilage to bone, characterized by distinct zonal organization with a cartilaginous core and calcified outer tissue [155]. In the same year, Tam et al. used hPSCs to successfully construct cartilaginous organoids in a similar way [156]. Hall et al. (2021) employed a bottom-up strategy to assemble osteochondral organoids by combining iPSC-derived cartilage microtissues (iCMTs) with pre-hypertrophic “callus organoids” (COs) from human periosteum-derived cells. The resulting biphasic constructs demonstrated zone-specific functionality, wherein the CO layer underwent mineralization and expressed osteogenic markers (e.g., osteocalcin), while the

iCMT layer maintained a stable cartilaginous phenotype (SOX9, COL2 positive), together recapitulating osteochondral tissue remodeling and integration upon in vivo implantation [157]. In 2022, Rodríguez et al. constructed osteochondral organoids derived from hPSCs overexpressing osteoprotegerin with an additional 19 amino acids at its C-terminus (osteoprotegerin extralong (OPG-XL)). The presence of OPG-XL in human cartilage and bone not only increases the formation of osteoclasts in response to RANKL but also directly influences the development of chondrocytes and osteoblasts, leading to mineralization of the extracellular matrix [13]. In 2023, Khan et al. proposed a systematic and targeted process for producing bone marrow organoids from hPSCs that are committed to mesenchymal, endothelial, and hematopoietic lineages [7].

The remarkable differentiation capacity of human pluripotent stem cells (hPSCs) has enabled the generation of complex, multicellular organoid models across multiple tissue types and disease domains, as summarized in Fig. 10. These platforms recapitulate key aspects of human organ development and pathophysiology, providing valuable tools for disease modeling and drug screening. For instance, guided differentiation protocols have been established to direct hPSCs toward cardiomyocyte lineages, facilitating the formation of cardiac organoids that mimic structural and functional traits of heart tissue (Fig. 10a) [158]. Similarly, kidney organoids with intricate glomerular and tubular structures have been constructed through staged morphogen exposure, modeling human nephrogenesis and polycystic kidney disease (Fig. 10b) [150]. In the skeletal domain, chondro-osteo-organoids have been engineered to emulate the bone-cartilage interface. The differentiation process is often enhanced by incorporating insulin to boost differentiation capacity, leveraging the IGF signaling pathway to promote robust matrix deposition and cellular zonation reminiscent of developing joints. (Fig. 10c–d) [157]. Most recently, hPSCs have also been leveraged to generate bone marrow organoids by co-differentiating mesenchymal, endothelial, and hematopoietic lineages in an integrated system, thereby modeling the hematopoietic niche and its associated malignancies (Fig. 10e) [7]. Together, these advances underscore the broad utility of hPSC-derived organoids in mimicking human tissue complexity and disease mechanisms.

Table 2 provides a consolidated summary of the studies discussed above, systematically comparing the key features of various bone organoid models. It encompasses not only the hPSC-based approaches detailed in this section but also includes representative models constructed directly from mesenchymal stem cells or terminal differentiated cells, such as osteoblasts and osteoclasts, offering a broader perspective on the field [22, 155, 159, 160].

Conclusion and Future Perspective

The development of functional bone organoids faces several technological challenges. First, the efficiency of the differentiation of hPSCs into osteoblasts is relatively low, and current methods still fall short in terms of purity and efficiency. Additionally, there is an urgent need to optimize scaffold materials and their physical/chemical stimuli to better adapt to hPSCs and simulate the in vivo environment. Furthermore, effectively integrating multiple cell types and their interactions remains a significant hurdle. To address these shortcomings, future studies should focus on the following areas: (1) enhancing the differentiation efficiency of hPSCs and utilizing artificial intelligence (AI) technologies to improve cell quality, optimizing scaffold materials for precise control of porosity and hardness to meet the specific requirements of bone tissue; (2) combining physical and chemical stimuli to increase cellular viability; (3) research into vascularization and mineralization; and (4) methods such as microfluidic channels and endothelial cell self-assembly that hold promise for supporting the construction of vascular networks. Concurrently, bioengineering approaches to replicate nanoscale mineral structures in bone are expected to yield significant advancements. Although the study of bone organoids is primarily at the basic experimental stage, there are extensive prospects for future clinical applications, including bone defect repair, personalized drug screening, and increased graft survival rates.

Challenges in Current Technology and Future Directions for Improvement

Stepwise and Xeno-Free in Vitro Induction of hPSCs into Osteoblast-Like Cells

The construction of bone organoids involves the use of osteogenic cells for building bone tissue, osteoclasts for bone remodeling, and vascular endothelial cells for transporting nutrients and oxygen. Although progress has been made in the differentiation of hPSCs into osteogenic cells, the differentiation of hPSCs into osteoclasts and endothelial cells is still a lengthy process.

The first challenge we face is that hPSCs encounter a series of difficulties during their reprogramming, culture, and differentiation processes. There are several shortcomings and improvements for hPSCs, specifically, as follows: (1) optimizing the induction process and investigating media components that reduce or eliminate the reliance on factors from serum [34, 161, 162]; (2) the overexpression of reprogramming factors by viruses must be replaced by safer and noninvasive means [163]; (3) hPSCs into osteoblasts require high-quality homogeneous cell populations without the risk

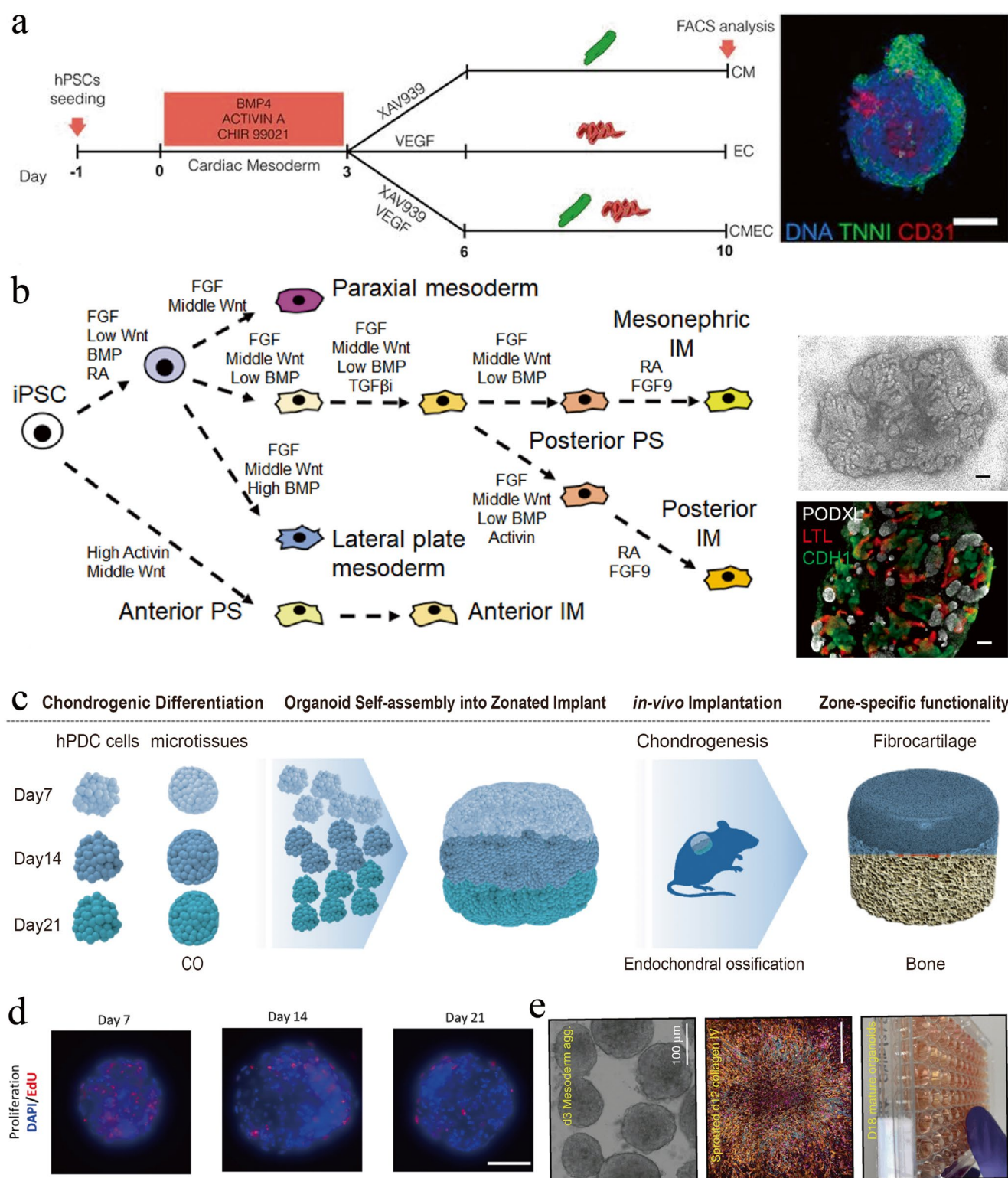


Fig. 10 hPSC-derived organoid manufacturing approaches have been utilized in several disease domains. **(a)** Differentiation protocol for determining cardiomyocyte cell fates and constructing organoids. Reproduced with permission [158]. Copyright 2017, The Company of Biologists Ltd. **(b)** Schematic illustration of the construction of complex kidney organoids with hPSCs. Reproduced with permission [150]. Copyright 2020, Cell Press. **(c-d)** Schematic illustrating the

structure of chondro-osteo-organoids generated from hPSCs. Reproduced with permission [157]. Copyright 2021, Elsevier. **(e)** Differentiation of hPSCs into mesenchymal, endothelial, and hematopoietic lineages for the manufacture of bone marrow organoids. Reproduced under terms of the CC-BY license [7]. Copyright 2023, Khan, Abdullah O; Rodriguez-Romera, Antonio, published by the American Association for Cancer Research

Table 2 Generation of related bone organs via hPSCs and primary cells

Cell type	Construction method	Organoid type	Ref.
hiPSCs	Encapsulation of mesenchymal progenitor cells derived from hiPSCs into a dual-flow bioreactor for chondro-osteo-tissue.	Chondro-osteo-organoid	[145]
Mouse iPSCs	Sequentially, TGF- β 3, doxycycline, and three osteogenesis factors were introduced in chronological order to accelerate the proliferation of individual cells for the development of an organoid.	Cartilaginous organoids	[155]
hiPSCs	The hiPSCs were directed toward the mesodermal lineage and underwent differentiation into chondrocytes, which then underwent self-assembly to form cartilaginous organoids.	Cartilaginous organoids	[156]
Human periosteum derived cells	Cartilage microstructures from two different sources are transformed into cartilage implants in specific areas.	Chondro-osteo-organoid	[157]
hiPSCs	Constructing osteochondral organoids using hiPSCs expressing osteoprotegerin.	Cartilaginous organoids	[13]
hiPSCs	Gradually differentiate hiPSCs into mesenchymal, endothelial, and hematopoietic lineages to manufacture bone marrow organoids.	Bone marrow organoids	[7]
Osteocytes	The osteocytes were encapsulated within the collagen matrix and the manufactured system to generate woven bone.	Woven bone organoid	[159]
Osteoblast Osteoclast	The model comprises a mixture of osteoblastic and osteoclastic cells introduced into the trabeculae of the femoral head.	Bone organoid	[155]
Rabbit MSCs	Preparing decellularized matrix of bone, and created a three-dimensional reconstruction of the blood arteries using PLLA.	Bone organoid	[22]
hMSCs	The objective is to create gelatin-based microcryogels that have been altered with HA and HYP in order to promote cartilage and osteogenic differentiation.	Osteochondral organoids	[160]

of residual teratogenic cells, which is further emphasized in the purification of osteoblasts (incomplete purification increases the number of unpurified hybrid cells, which may result in a decrease in the efficiency of final osteogenic differentiation, and epigenetic variation in cell culture would lead to a greater risk of tumorigenesis) [164]; and (4) a lack of reliable assays to detect the cellular stress caused by poor growth circumstances and excessive apoptotic and spontaneous differentiation signals during cell processing [165]; and (5) current approaches for maintaining undifferentiated hPSCs rely on feeder cells or an extracellular matrix [166]. A better understanding of how chemical moieties support or direct biological action will lead to improvements in and the development of new synthetic substrates [167]. (6) Although the development of hPSC technology may overcome ethical barriers to the use of embryos in research, issues remain regarding the biological equivalency and safety of hPSC lines in comparison with those of hESCs [132]. Some issues include limited or reduced developmental capacity, as well as unpredictability in the beginning cell type employed for reprogramming [168].

The process of hiPSC-induced osteogenesis *in vitro* is crucial for the construction of bone organoids and hinges on the development of direct induction systems under

defined conditions. The protocol needs to ensure the safety and convenience of generating osteoblast-like cells to fulfill the demands of bone engineering. As mentioned above, several protocols, including the EB induction method and monolayer induction method, have made progress in obtaining osteoblasts from hPSCs. However, it is evident that the reported methods are not without imperfections and require further refinement [169]. Key future directions include establishing xeno-free induction systems and improving differentiation efficiency. Although current methods can produce osteoblast-like cells, challenges such as low differentiation efficiency, cell heterogeneity, and difficulties in obtaining high-purity and functional osteoblasts persist. Additionally, the quality and stability of hPSCs are affected by factors such as cellular mutations and epigenetic changes. A major issue is the unclear mechanisms of growth factors at different induction stages, which can lead to inefficiencies. As a result, continuous refinement of additives such as BMPs, Wnt, and FGF is necessary, along with optimizing their dosage and timing [170, 171]. Advances in multimodal spatiotemporal transcriptomics are expected to provide insights into the molecular mechanisms of osteogenic differentiation. Future efforts should focus on refining culture media and improving the regulation of signaling

pathways to increase differentiation efficiency and reduce heterogeneity, thus enabling the successful generation of osteoblast-like cells from hPSCs.

Multiple hPSC lines, such as MCND-TENS2, GM28404*B, and others, have been successfully induced to differentiate into osteoclasts [59, 65]. However, the current systematic and comprehensive efforts to evaluate which specific hPSCs exhibit greater differentiation efficiency are currently limited. Only one study reported that there was no significant difference in osteoclast differentiation efficiency between PBMCs and fibroblast-derived iPSC lines [69]. Owing to the limited quantity of PBMCs collected from the human body and their difficulty in expanding *in vitro* [172], the use of hPSCs for osteoclast differentiation has broad clinical potential. However, the current protocols for hPSCs to differentiate into osteoclasts are still relatively limited. Most experimental protocols focus mainly on the terminal differentiation of osteoclasts after EB suspension culture or the mesodermal stage. However, there are relatively few experimental protocols for differentiating hPSCs into osteoclasts through monolayer culture. Some studies suggest that differentiation on the basis of monolayer culture results in the production of fewer hematopoietic cells, mainly CD34⁺ cells, which subsequently do not differentiate into osteoclasts [173]. The majority of commercially available osteoclast induction media contain animal-derived components such as FBS, and a considerable amount of time is needed to develop well-defined chemical compositions. Only Sana Ansari et al. proposed a serum-free substitute medium with a precisely defined chemical composition for the osteoclast differentiation of PBMCs [174]. Currently, there is no well-defined medium for inducing the differentiation of hPSCs into osteoclasts, making the optimization of culture medium components a challenging task for enhancing the efficiency of hPSC differentiation into osteoclasts. For the induction of endothelial-like cells, challenges such as rapid dedifferentiation, loss of identifiable features, low induction efficiency, and complex culture conditions exist. Notably, it is also difficult to purify target cells from cardiovascular progenitor cells. Therefore, achieving multiple functionalities remains a crucial challenge in the development of integrated bone organoids. The complex interactions between different cell types make direct coculture difficult to control because of differences in the direction of differentiation. Additionally, further studies are needed to elucidate lineage specificity, which enables the differentiation of functionally distinct endothelial cells customized for their specific destinations [175]. Currently, the ability of commonly used stem cells to differentiate into endothelial-like cells is extremely limited, as they are unable to meet the demand for the large number of cells required for vascular formation. In future research, how to stagewise add endothelial-like factors,

relevant growth factors, and biomaterials during the cultivation process to promote induced differentiation should be explored. Furthermore, researchers may attempt to integrate organoid systems into the host vascular system to increase the relative stability of the extracellular environment [6]. With respect to the issues in the aforementioned induction and differentiation process, we have identified the following future directions for development.

In recent years, the rapid development of deep learning has opened new directions for bone tissue engineering. For example, deep learning models have been applied to screen small molecules and peptides that promote osteogenic differentiation [176]. Computer vision technology has also been used to classify, evaluate, and predict stem cell differentiation and bone formation processes [177]. Furthermore, AI technology is revolutionizing the field of biomaterials design, optimizing the mechanical strength, porosity, and biodegradability of hydrogels [178].

With the continuous development of AI technologies and the decreasing costs of omics techniques, such as RNA sequencing (RNA-seq) [179] and chromatin immunoprecipitation sequencing (ChIP-seq) [180], the development of osteogenic differentiation systems has been further promoted, providing multidimensional data sources to elucidate the mechanisms of osteogenic differentiation. For example, Shen et al. used Seurat for data preprocessing (normalization and dimensionality reduction through PCA and UMAP) and the Leiden algorithm for clustering to identify MSC subgroups with different osteogenic potentials. They then applied a random walk algorithm on the protein-protein interaction (PPI) network to screen genes with topological structures similar to osteogenesis-specific genes (e.g., FOXA1) and validated their role in regulating osteogenesis [181]. Bandyopadhyay et al. used spatial transcriptomics and multi-ion beam imaging (MIBI) to map MSC subgroups in human bone marrow and found that osteoprogenitors aggregate near blood vessels. They applied graph convolutional networks (GCNs) to model the spatial interactions between mesenchymal stem cells and their microenvironment, predicting that hypoxia (via HIF-1 α) enhances osteogenic differentiation [182]. Feng, by using support vector machine (SVM) and random forest analysis of proteomic data from osteoporosis patients, identified 205 differentially expressed proteins enriched in extracellular matrix (ECM) pathways. The study found that the ECM-receptor interaction pathway is a central regulatory factor for osteogenic differentiation, providing targets for therapeutic intervention [183].

The combination of AI and omics technologies enables researchers to analyze stem cell gene expression, proteomics, and epigenetic features in greater detail. It allows for early, non-invasive predictions of osteogenic differentiation and high-throughput analysis of complex data, such

as morphology and omics. As AI and omics technologies continue to integrate, bone organoid research will become more precise and efficient, providing significant support for personalized therapy and the future development of bone tissue engineering [184].

However, there are still significant challenges to address. Currently, there is no unified quantitative system for optimizing cell digestion methods to obtain osteoblast-like cells, primarily due to the low survival rates associated with pancreatic enzymes and the difficulty in accurately quantifying variables such as digestion time and temperature. Therefore, future studies should leverage advanced techniques such as single-cell transcriptome sequencing to better understand the interplay between the physiological state of cells and chemical factors. Additionally, research should focus on establishing a quantitative evaluation system using indicators such as cell viability, morphological changes, and surface marker variations to assess digestion effectiveness. Optimizing enzyme types, solution composition, and digestion conditions is crucial, and standardized procedures should be implemented to improve the reproducibility and comparability of research findings.

Optimization of Scaffold Materials and Physical/Chemical Stimuli

Recent advancements in biomaterials and physical stimulation have shown great potential in promoting hPSC differentiation toward osteogenesis, yet challenges remain in optimizing scaffold design and stimuli to meet the specific needs of hPSCs.

The use of scaffolding materials and extracellular matrices supports hPSC differentiation toward osteogenesis, although most scaffolds used were initially designed for hMSCs. Examples include PCL/PVDF [185], hydrogels [186], and nanofibers [187]. However, these materials often overlook the unique survival and differentiation processes of hPSCs. Future research should focus on creating scaffold materials specifically tailored to hPSCs and optimizing culture environments. Investigating the effects of multiple factors on scaffold properties and osteogenic outcomes, such as the use of chemical topography chips, represents a promising research direction.

In addition, chemical stimuli such as growth factors, chemical small molecules, and hormones have been widely utilized, but their short half-lives are prone to losing their activities [188]. In contrast, physical stimuli, including magnetic [189], electrical [190], and mechanical stimuli [191]; oxygen concentration [192]; intercellular shear stress [193]; and sound and light [194, 195], as well as the activation of signaling pathways, modulation of ion channels, and modulation of bone-related gene expression, can continuously

increase and maintain cellular activity [196]. However, inadequate physical stimulation, such as microgravity, can inhibit osteoblast differentiation [197]. Therefore, physical stimulation should be used to minimize the damage caused by adverse external factors to osteoblasts.

Differentiation and Interaction of Multicellular Types

The lack of research on cell–cell interactions has also hampered the progress of bone organoid construction. Currently, the construction of bone organoids focuses primarily on the skeletal system, and sufficient consideration of the coordinated interactions among multiple systems is lacking. However, the healing of bone defects in the human body involves complex regulation of various systems, including the nervous, endocrine, and immune systems. Therefore, a more comprehensive consideration of the interactions among multiple systems is needed in the construction of bone organoids. However, research on cell–cell interactions faces technical challenges. Despite the availability of numerous biophotonic imaging and cell tracking techniques, accurately monitoring and quantifying cell–cell signaling and interactions remains a challenge. Overcoming these technical limitations is necessary for better understanding and simulating cell–cell interactions, thereby advancing the field of bone organ construction.

Challenges in Bone Organoid Engineering: Overcoming Maturity, Vascularization, and Complexity

Currently, several challenges in the process of building bone organoids urgently need to be addressed.

First, one of the key challenges of hPSC-derived organoids is their maturity and scalability. Owing to limited culture conditions (such as the supply of oxygen and nutrients to the core of the organoid), hPSCs can only grow to the micron- to millimeter-scale level, thus limiting the lifespan and function of the organoid [198]. Although this is challenging, scientists are continually striving to overcome this issue and have made some progress. For example, the integration of emerging 3D printing technologies or cell-based bottom-up manufacturing enables the construction of biomimetic and layered structures, expanding the size range of organoids [199].

Second, additional vascular endothelial cells can be integrated into the bone organoid culture system to achieve vascularization. A simple way to vascularize bone organoids is to provide endothelial cells with a variety of chemical, cellular or biophysical cues and allow them to self-assemble into vascular networks; however, this method uses the tube-forming ability of endothelial cells, and the resulting

microvascular system is variable and may not always reflect physiological conditions, often necessitating *in vivo* transplantation for perfusion. The rationale for this self-assembly stems from the inherent developmental program of organoids, which naturally recapitulate *in vivo* neo-angiogenesis mechanisms. As organoids increase in size, they develop a hypoxic core due to limited oxygen and nutrient diffusion [85, 200]. This hypoxic microenvironment is a critical regulator that triggers angiogenesis, primarily through the stabilization of the transcription factor HIF-1 α (Hypoxia-Inducible Factor 1- α) [85, 201]. HIF-1 α , in turn, activates the transcription of key pro-angiogenic factors, most notably Vascular Endothelial Growth Factor (VEGF), to recruit and organize new blood vessel formation [202]. The Insulin-like Growth Factor (IGF) pathway is a crucial modulator of this specific mechanism. IGF signaling (via both IGF-1 and IGF-2) has been shown to potently stimulate and enhance the activity and nuclear translocation of HIF-1 α [203]. This creates a powerful synergistic loop within the organoid developmental phase: the hypoxic environment stabilizes HIF-1 α , and simultaneously, IGF signaling amplifies HIF-1 α 's transcriptional activity. This synergy results in a robust upregulation of VEGF, which is the central driver for endothelial cell migration and tube formation, thereby promoting the vital neo-angiogenesis required for organoid survival and maturation [204].

In addition to the aforementioned challenges related to the bone organoids themselves, another difficulty lies in the safety concerns associated with hiPSCs. First, there is the risk of genomic instability, as mutations may accumulate during prolonged culture, increasing the potential for tumor formation [205, 206]. Undifferentiated hiPSCs or their derivatives may exhibit a higher tumorigenic potential, necessitating effective techniques to screen and eliminate undifferentiated cells [207]. Second, allogeneic hiPSC transplantation may trigger immune rejection. Although autologous hiPSCs reduce immune response, allogeneic cells may still be recognized and attacked by the host's immune system [208, 209]. Furthermore, the generation of hiPSCs often relies on viral vector transfection, which may carry the risk of insertional mutagenesis, further exacerbating genomic instability [210, 211]. Incomplete or abnormal differentiation processes could also result in unstable cell populations, potentially leading to tumors or loss of function. Therefore, precise differentiation control and regulation of cell proliferation cycles are crucial to ensuring safety in the construction of bone organoids. Additionally, the post-transplant microenvironment significantly affects the survival and functionality of hiPSCs. Inadequate substrates or oxygen and nutrient supply may lead to cell death or increased cancer risk [212, 213]. Furthermore, with changes in immune tolerance, long-term use of hiPSCs may

also result in immune rejection. Lastly, contamination of cell cultures presents another potential safety hazard [214]. Despite these challenges, advances in gene editing technologies, virus-free transfection methods, and more precise differentiation induction strategies offer hope for overcoming the safety concerns associated with hiPSCs, thus paving the way for their broader clinical application [215].

Finally, the unique structural and functional complexities of bone organoids necessitate innovative approaches to address their stiffness variability, calcification, and mineralization challenges, ultimately increasing their maturity and regenerative potential. The construction of bone organs is different from that of other mature soft tissue organoids at present. Bone organoids contain a variety of soft (bone marrow) and hard (mineralized bone) organ components, the stiffness requirements of the cells are heterogeneous, and researchers may need to construct at least two complex environments with different stiffness levels at the same time. Therefore, researchers still need to explore the use of different cell types and tissue engineering methods to increase the maturity and functionality of organoids, addressing the growth limitations of these structures [216]. Like natural bones, calcification and mineralization are crucial in the process of bone organogenesis. The existing bone organoid models fail to recapitulate the nanoscale calcification of bone, its inherent complexity, or the intricate interactions between cells and both the matrix and minerals. Consequently, crafting bioengineered bone microenvironments with profound biological complexity and potent osteoregenerative capabilities remains a formidable challenge within the domain of bone organoid engineering [198].

Perspectives on the Clinical Applications of Bone Organoids

Research on bone organoids is still limited to basic medical experiments. Further research should be conducted to provide additional perspectives and advancements for the future construction of bone organoids, including all *in vivo* bone tissue cells, which are thus capable of bone remodeling. In the future, bone organoids will gradually transition from basic research to clinical applications, offering endless opportunities for the treatment of human orthopedic diseases.

In clinical orthopedics, especially in the face of external trauma, the construction and implantation of bone defect organoid models will greatly assist clinicians in repairing large bone defects. Compared with traditional mechanical tissue engineering scaffolds, bone organoids derived from human cells possess better biocompatibility, reducing the risk of rejection. In the field of osteoarthritis, with the incorporation of immune cells into stem cell niches during the

process of bone organoid construction, the future implantation of bone organs may gradually replace traditional cell therapies. Additionally, the addition of inorganic components such as calcium or phosphorus and the programmed mineralization process influence the maturity and functionality of bone organoids. Hyaluronic acid and other biomaterials hold promise for fundamentally transforming the field of orthopedic regeneration. In the near future, artificial bone grafts will be used for in vitro cultivation for retransplantation to maximize cell viability and promote osteogenic differentiation.

Furthermore, in the field of basic medical research, future researchers will further address the aforementioned issues by constructing standardized bone organoid models. Leveraging the advantages of bone organoids having physicochemical properties similar to or identical to those of in vivo hydrogels, this approach can shorten screening periods and provide more personalized treatments for drug toxicity analysis and drug screening. Additionally, by simulating the real in vivo environment and replicating cell communication, bone organoid models can more effectively evaluate implanted materials.

In summary, in the future, bone organoids will make significant contributions to the flourishing development of human medicine because of their limited potential.

Abbreviations

AA	Ascorbic Acid	CTR	Calcitonin Receptor
AI	Artificial Intelligence	DLP	Digital Light Processing
AKT/mTOR/p70S6K	Protein Kinase B/mechanistic Target of Rapamycin/ribosomal protein S6 kinase	DNA	Deoxyribonucleic Acid
ALP	Alkaline Phosphatase	Dex	Dexamethasone
ALPL	Alkaline Phosphatase, Liver/Bone/Kidney	EB	Embryoid Body
BCoc	Bioactive Ceramic-reinforced Composite	ECM	Extracellular Matrix
BG	Beta-Glycerophosphate	EPCs	Endothelial Progenitor Cells
BMM	Bone Marrow Mononuclear	ERK	Extracellular Signal-Regulated Kinase
BMP-SMAD	Bone Morphogenetic Protein-Small Mothers Against Decapentaplegic	FACS	Fluorescence-Activated Cell Sorting
BMP2	Bone Morphogenetic Protein 2	FBS	Fetal Bovine Serum
BMP4	Bone Morphogenetic Protein 4	FGF	Fibroblast Growth Factor
BMSCs	Bone Marrow-derived Mesenchymal Stem Cells	FLK-1	Fetal Liver Kinase 1
C-FMS	Colony stimulating factor 1 receptor	HA	Hyaluronic Acid
c-Kit	Mast/stem cell growth factor receptor Kit	HSCs	Hematopoietic Stem Cells
CHIR	CHIR99021	HYP	Hydroxyproline
CMP	Common Myeloid Progenitor	IFITM5	Interferon Induced Transmembrane Protein 5
COL1A1	Collagen Type I Alpha 1 Chain	IL-3R	Interleukin-3 Receptor
		JAK/STAT	Janus Kinase/Signal Transducer and Activator of Transcription
		LDL	Low-Density Lipoprotein
		M-CSF	Macrophage Colony-Stimulating Factor
		MACS	Magnetic-Activated Cell Sorting
		MAPK	Mitogen-Activated Protein Kinase
		MSCs	Mesenchymal Stem Cells
		NEAA	Non-Essential Amino Acids
		OCN	Osteocalcin
		ONN	Osteonectin
		OPG-XL	Osteoprotegerin extra-long
		OSCAR	Osteoclast Associated Ig-Like Receptor
		PBMCs	Peripheral Blood Mononuclear Cell
		PCL	Polycaprolactone
		PDGF-BB	Platelet-Derived Growth Factor-BB
		PI3K	Phosphoinositide 3-Kinase
		PKD	Polycystic Kidney Disease
		PLG	Plasminogen
		PLLA	Poly-L-lactic Acid
		PTH	Parathyroid Hormone
		RA	Retinoic Acid
		RANK	Receptor Activator of Nuclear Factor Kappa-B
		RANKL	Receptor Activator of Nuclear Factor Kappa-B Ligand
		RhoA/ROCK	Ras homolog gene family member A/Rho-associated protein kinase
		RUNX2	Runt-related transcription factor 2
		TGF- β	Transforming Growth Factor-beta

TRAP	Tartrate-Resistant Acid Phosphatase
VEGF	Vascular Endothelial Growth Factor
hESCs	human Embryonic Stem Cells
hMSCs	human Mesenchymal Stem Cells
hPDCs	human Periosteum Derived Cells
hPSC-MSCs	human Pluripotent Stem Cell-derived Mesenchymal Stem Cells
hPSCs	human Pluripotent Stem Cells
hiPSCs	human induced Pluripotent Stem Cells
sRANK	soluble Receptor Activator of Nuclear Factor Kappa-B
vWF	von Willebrand Factor
$\alpha v\beta 3$	integrin alpha V beta 3
3D	Three-dimensional

Acknowledgements The authors declare that they have not use AI-generated work in this manuscript.

Author Contributions P.Z., R.Z. and C.Z. contributed to critically reviewing the manuscript. Z.X., M.L. and S.W. wrote the main manuscript text and prepared Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10; Table 1, and 2, TM, HZ, YG, YS, YK, LQ contributed to the draft and review of the manuscript. All the authors have reviewed and approved the article and have done their due diligence to ensure the integrity of the manuscript.

Funding This research received funding from National Natural Science Foundation of China Regional Program (No. 82560331), the Natural Science Foundation of Gansu Province (No. 25JRRA1236, 24JRRA945, 24JRRA948), the Fundamental Research Funds for the Central Universities (Izujbky-2025-30), the Gansu Province Health Industry Research Project (No. GSWSKY2023-71, GSWSQNPY2025-19), the CSA West China Clinical Research Fund (No. CSA-W2024-11), the Lanzhou University Hospital of Stomatology Research Support Fund (No. Izukqky-2023-t07) and Lanzhou Science and Technology Project(No.2025-2-151).

Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

Clinical trial number Not applicable.

Competing interests The authors declare no competing interests.

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